CR23532

CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS - II

B.E. CSE - Cyber Security

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Method note. The case facts supplied in the assignment are limited. Any event times or cross-source events not stated in the brief are explicitly marked as hypothetical. The procedures describe a proposed forensic workflow; they do not claim that a forensic acquisition or software examination was actually performed. The flowcharts are drawn to reflect the correct sequence of the answer, and common sources of error are discussed in the text rather than inserted as hidden procedural mistakes.

Evidence-handling principle. Preserve originals, work from verified copies, record provenance, normalize timestamps without overwriting raw values, validate important artifacts independently, and separate an account/device association from proof of an individual's conduct.

Assignment 2(1) - Suspected Unauthorized Access to a College Server

Questions 1-7

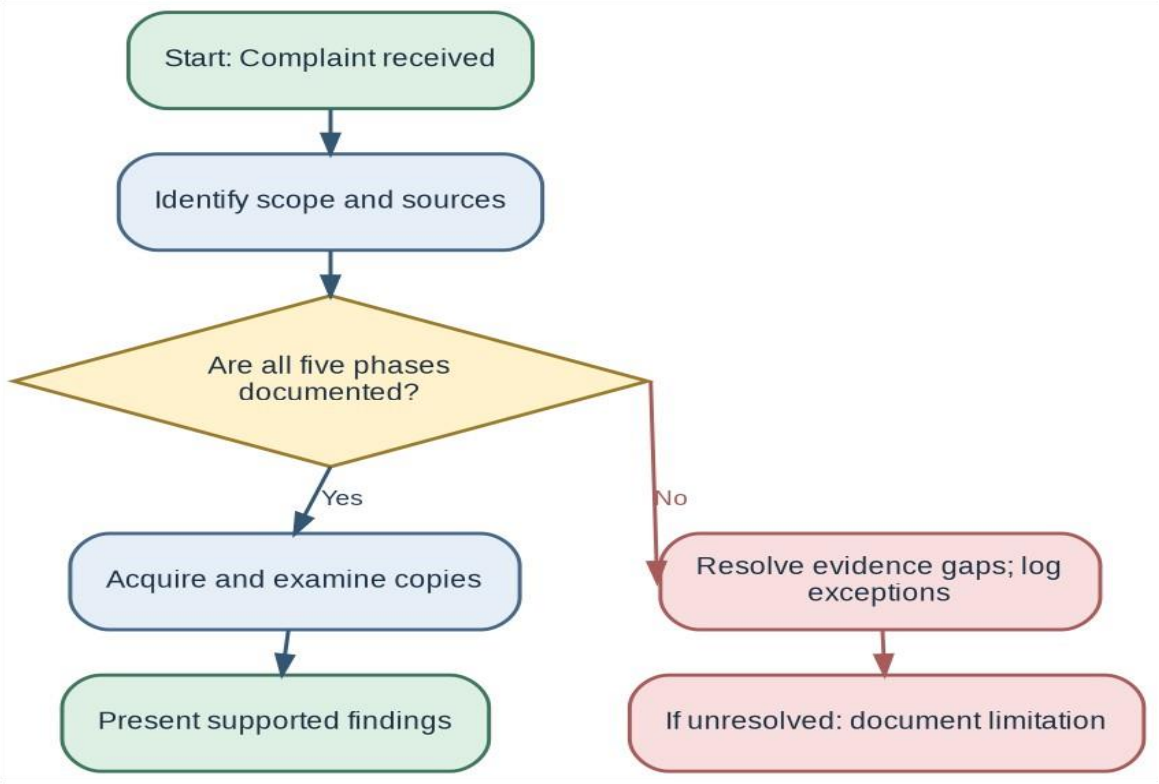
A college administrator reports that confidential examination-related files may have been accessed and copied outside normal working hours. The known sources are the complaint, a suspect workstation, server access logs, a USB device, and relevant files. The answers below treat the incident as an allegation to be tested, not as proof that a named person committed it.

1

Explain how the forensic team should move through the Identification, Collection, Examination, Analysis, and Presentation phases for this case.

Identification defines the incident scope, systems, accounts, relevant time window, file names, likely data paths, and people authorized to approve the investigation. Investigators record the complaint and identify the server, suspect workstation, USB device, access logs, and relevant files. They also identify volatile sources that may expire, such as active sessions or short-retention logs.

During Collection, preserve the original devices and logs, use a validated acquisition method and write blocker where appropriate, record serial numbers and custody transfers, and calculate cryptographic hashes. Examination works from verified forensic copies: recover file-system artifacts, inspect event and access logs, search for target files, and extract USB and user-activity artifacts. Analysis correlates timestamps, identities, paths, hashes, and system events to test competing explanations. Presentation reports the methods, limitations, supporting evidence, and conclusions in clear language. These phases may be revisited when new leads appear.



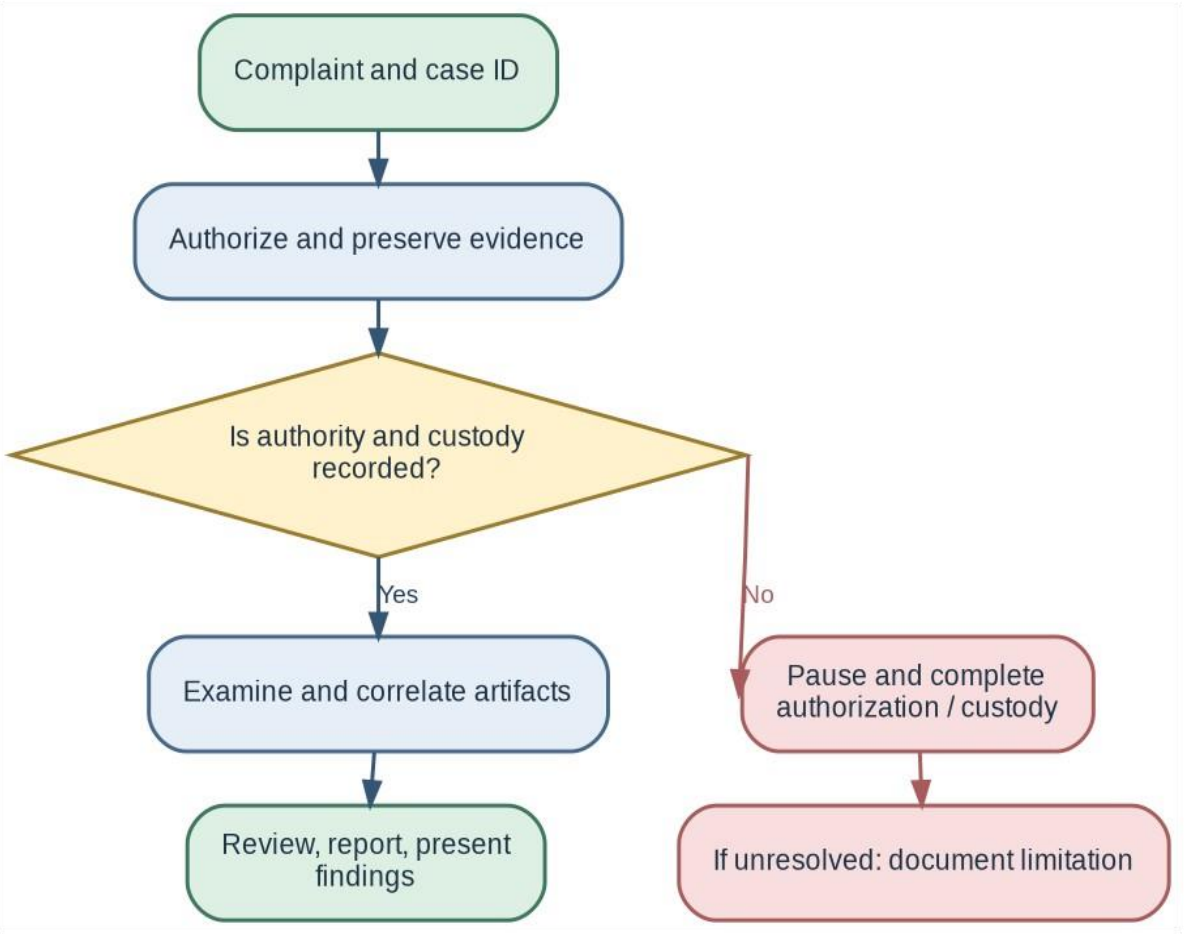
Flowchart 1. Question-specific reasoning sequence.

2

Prepare a forensic investigation flowchart showing the sequence of activities from the initial complaint to presentation of findings.

The sequence begins by registering the complaint, assigning a case identifier, and recording the allegation without assuming it is true. The investigator obtains written authority defining the systems, accounts, data categories, and time period in scope. Evidence is photographed, labeled, collected, and transferred under documented custody.
Original media are protected and forensic copies are verified before detailed examination.

Investigators then examine server logs, workstation artifacts, USB contents, authentication records, and target files. Relevant events are normalized to a common time reference and correlated. If a conflict or gap appears, it becomes a lead for a targeted, authorized follow-up rather than a reason to alter the original evidence. Findings are peer-reviewed where practicable, and a report explains what the evidence supports, what remains uncertain, and how the conclusion was reached.



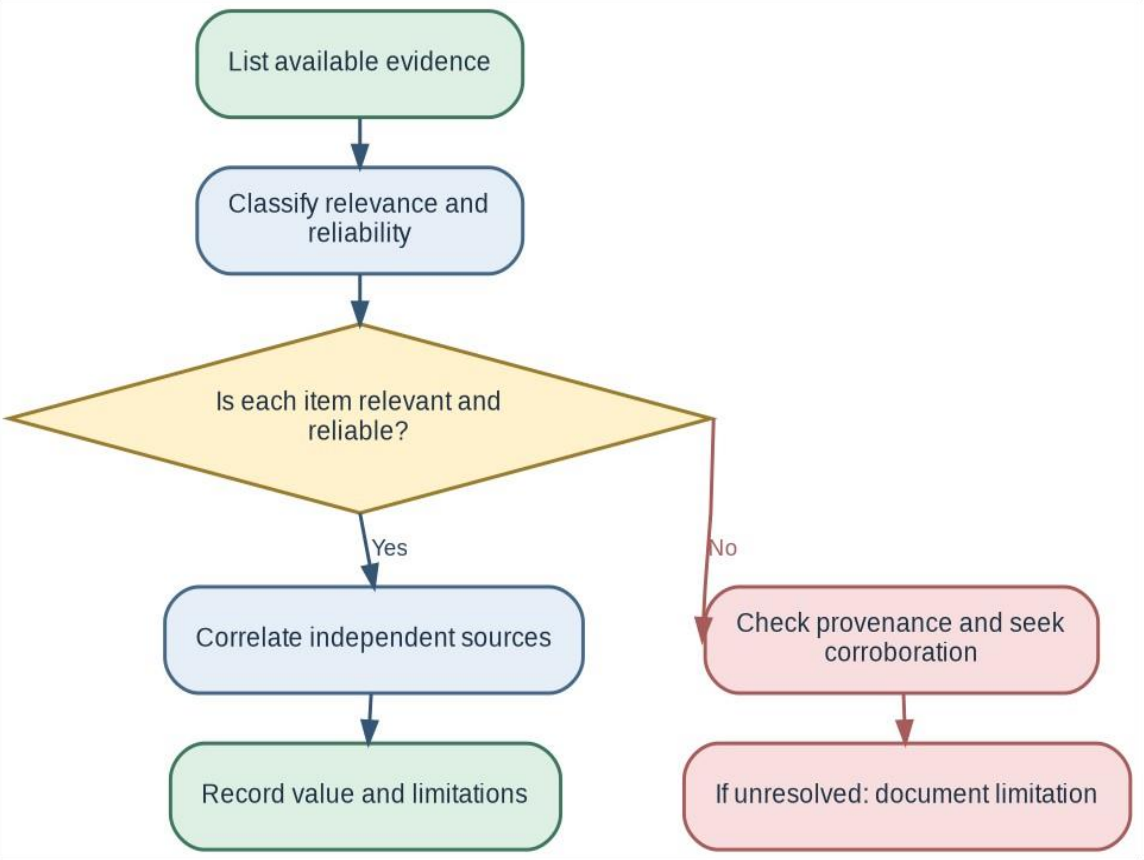
Flowchart 2. Question-specific reasoning sequence.

3

Identify at least five items of potential digital evidence in the case and classify the evidence according to its investigative value.

The most useful initial items are the administrator's complaint, suspect workstation, server access logs, recovered USB device, and the examination-related files. Additional sources to request under authorization include authentication logs, endpoint security events, firewall or proxy records, backup records, and server audit logs.
Evidence value depends on relevance, reliability, provenance, and how directly it addresses a disputed fact; it is not determined by the item alone.

Server-side records showing a successful authenticated read or download of a specific file can be highly probative of an account and operation. A USB device with a matching file hash can corroborate copying, but its presence alone does not prove who copied the file. A complaint provides context and directs collection, while a file's metadata may be supportive but can be changed. Investigators should preserve contradictory evidence and document both the evidentiary value and the limitations of each item.



Flowchart 3. Question-specific reasoning sequence.

Potential evidence and investigative value

Evidence item	Potential value	Classification / limitation
Administrator complaint	Establishes allegation, discovery time, affected files, and initial scope.	Contextual lead; statement should be checked against technical records.
Server access / audit logs	May record account, source address, session, object path, action, and time.	Potentially direct for a logged account operation; depends on audit completeness and integrity.
Suspect workstation forensic image	May contain logons, file access traces, browser records, program execution artifacts, and USB history.	Broad corroborative source; shared use and retention gaps limit attribution.

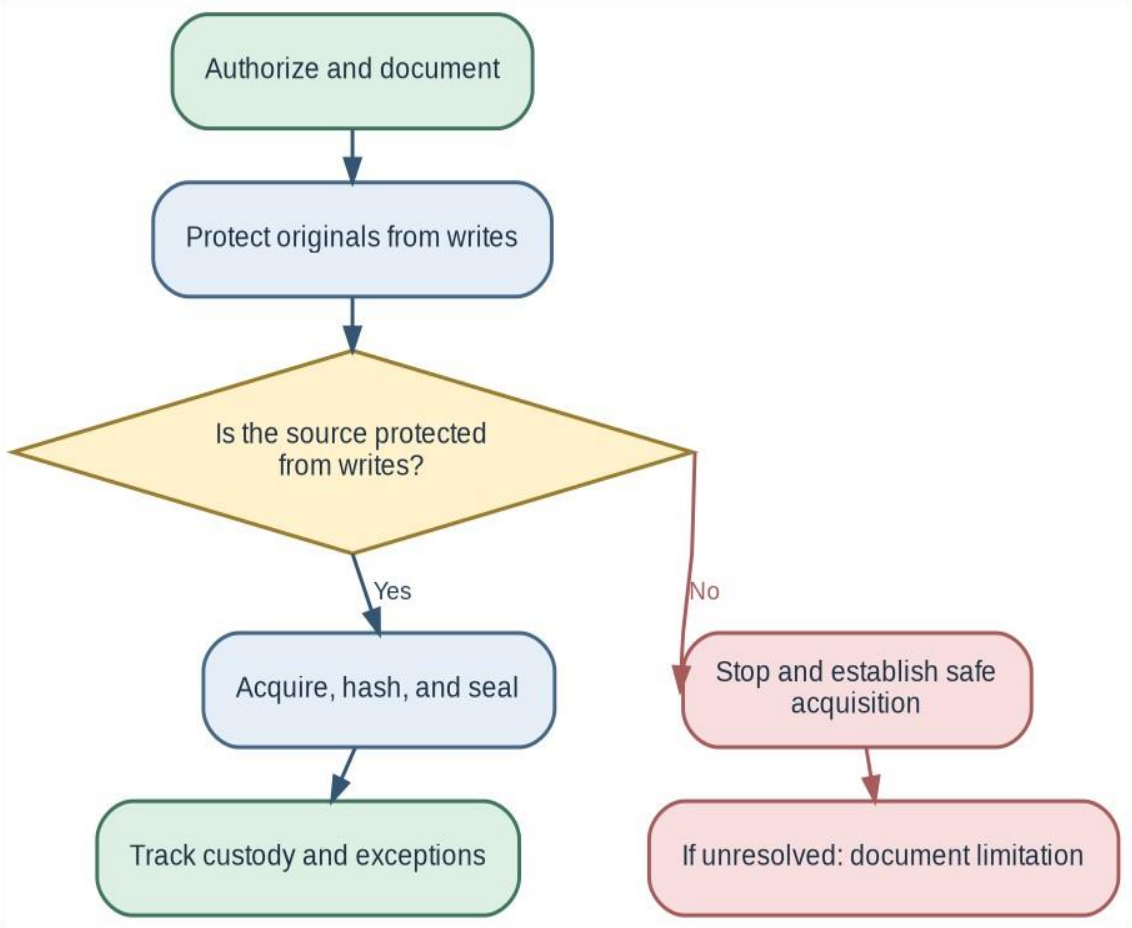
Evidence item	Potential value	Classification / limitation
Recovered USB device	May contain copied examination files, matching hashes, labels, or filesystem metadata.	Physical possession alone does not prove who copied or used the data.
Relevant examination files	Can identify target content, versions, metadata, and hashes for comparison.	Content/hash comparison supports identity; metadata may be altered or copied.
Authentication / endpoint logs (request if available)	May link a login session to a device, MFA event, process, or security alert.	Independent corroboration; requires authorized collection and clock normalization.
Network / proxy / firewall logs (request if available)	May corroborate data transfer destination, time, volume, or source IP.	Contextual; NAT, VPN, shared IPs, and incomplete logs can mislead.

4

Explain the precautions required during collection and preservation so that the evidence remains reliable and defensible.

Obtain proper authority before searching institutional systems or personal data, define scope, and use trained personnel. Photograph and document device connections, power state, date/time, labels, and visible identifiers before handling. Avoid browsing the suspect system normally. Decide whether live acquisition is necessary before powering down, because shutdown may destroy volatile evidence or make an encrypted volume inaccessible; any live response must be justified and logged.

Use a hardware write blocker for compatible storage acquisition, validated tools, clean destination media, and a documented procedure. Record the source and destination identifiers, tool and version, settings, start/end times, errors, and hashes. Seal and label originals, protect devices from physical damage, and maintain a chain-of-custody record for every transfer. Store working copies separately, restrict access, and never make unexplained changes to the only copy. If a tool reports read errors or an acquisition is incomplete, record the exception rather than concealing it.



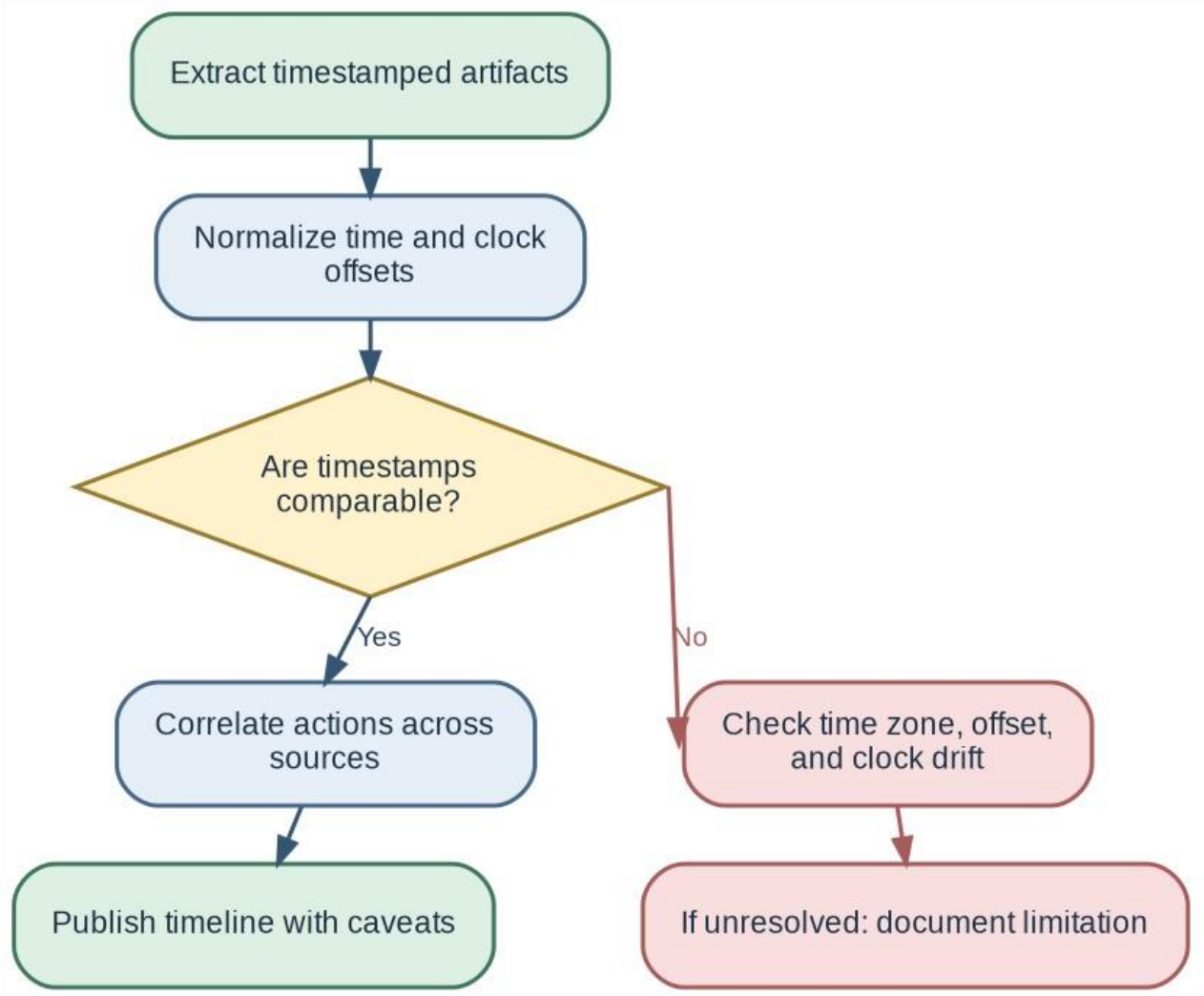
Flowchart 4. Question-specific reasoning sequence.

5

Describe how investigators can establish a timeline of the suspected unauthorized access using available evidence.

Build a timeline by extracting server authentication and file-access events, workstation logon and execution artifacts, USB connection events, file-system timestamps, relevant browser or application records, and any available network logs. Preserve each source timestamp exactly as reported, note the system's configured time zone and daylight-saving behavior, and estimate clock skew using reliable reference events. Convert copies of the timestamps to one chosen reference such as UTC while retaining the original values.

For each event, record source, event time, collection time, account, action, object or path, and confidence. A hypothetical sequence might be: successful account sign-in; access to a target file; USB insertion; creation of a copy on removable media; then logoff. Each link must be supported by artifacts rather than inferred from chronology alone. Distinguish a file's created, modified, accessed, and metadata-changed times; none by itself proves a person performed the action. Explain gaps and competing timelines.



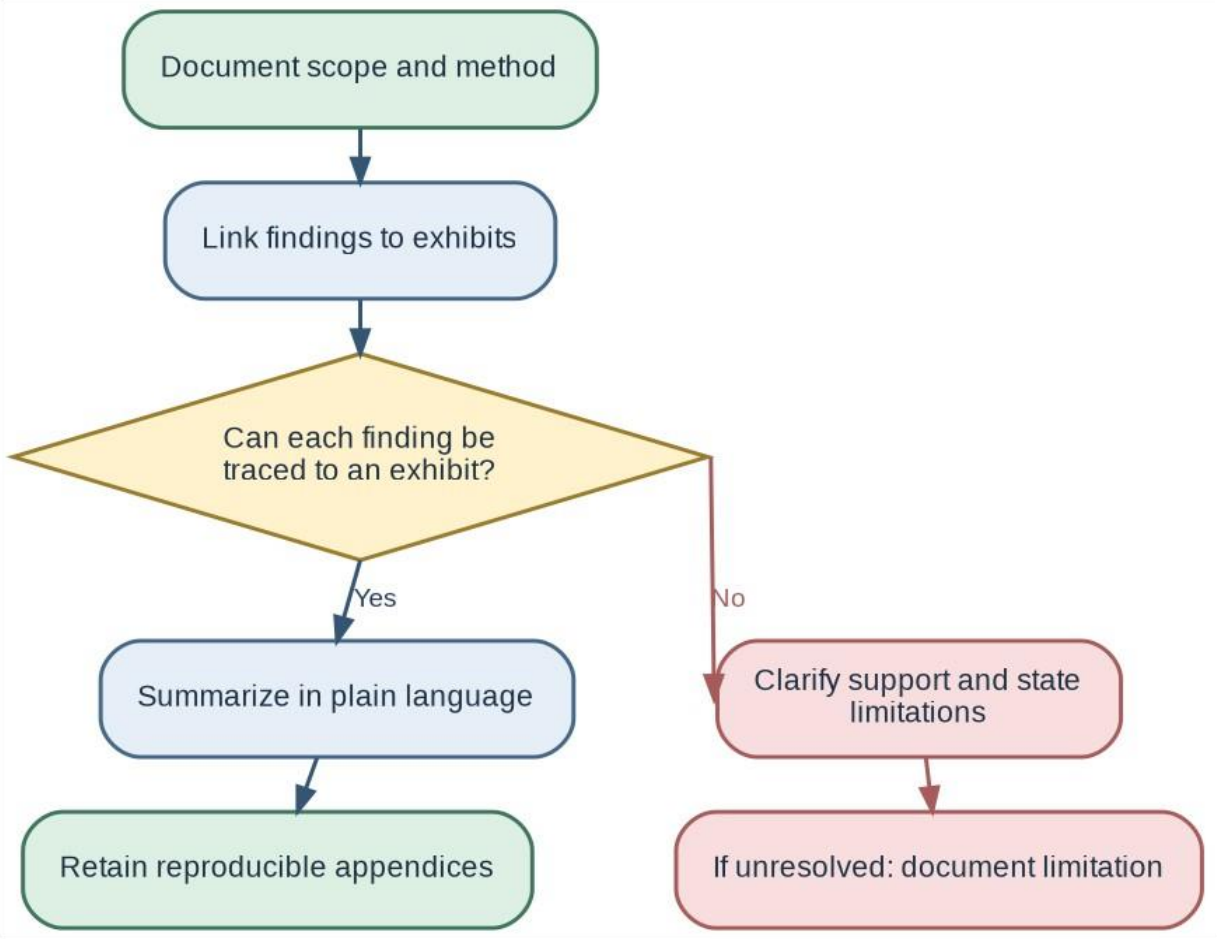
Flowchart 5. Question-specific reasoning sequence.

6

Explain what should be included in the final forensic report and how findings should be presented to a non-technical authority.

The report should have a case identifier; request and scope; authority and constraints; examiner details; evidence inventory and chain of custody; acquisition methods; tool names and versions; hash values; examination steps; findings linked to exhibits; timeline; limitations; and conclusion. It should distinguish observed facts, interpretations, assumptions, and unresolved questions. Negative findings are also reported when relevant, such as no matching file being found in a specified image or the absence of logs for a particular period.

For non-technical decision-makers, begin with a short executive summary that answers what was examined, what was established, how confident the examiner is, and what could not be established. Define terms such as hash, forensic image, and timestamp. Use a simple timeline and evidence table instead of raw log dumps, but retain technical appendices so another examiner can reproduce the analysis. Avoid claiming that a user personally performed an act merely because their account or device appears in an artifact.



Flowchart 6. Question-specific reasoning sequence.

Sample forensic report structure

Section	Required contents
1. Case summary	Case ID, authority, allegation, scope, key conclusion, and high-level limitations.
2. Evidence inventory	Unique IDs, source description, serials, condition, received date/time, seals, and custody references.
3. Methods	Acquisition and analysis procedures, write protection, tool/version, settings, hashes, time normalization, and validation.

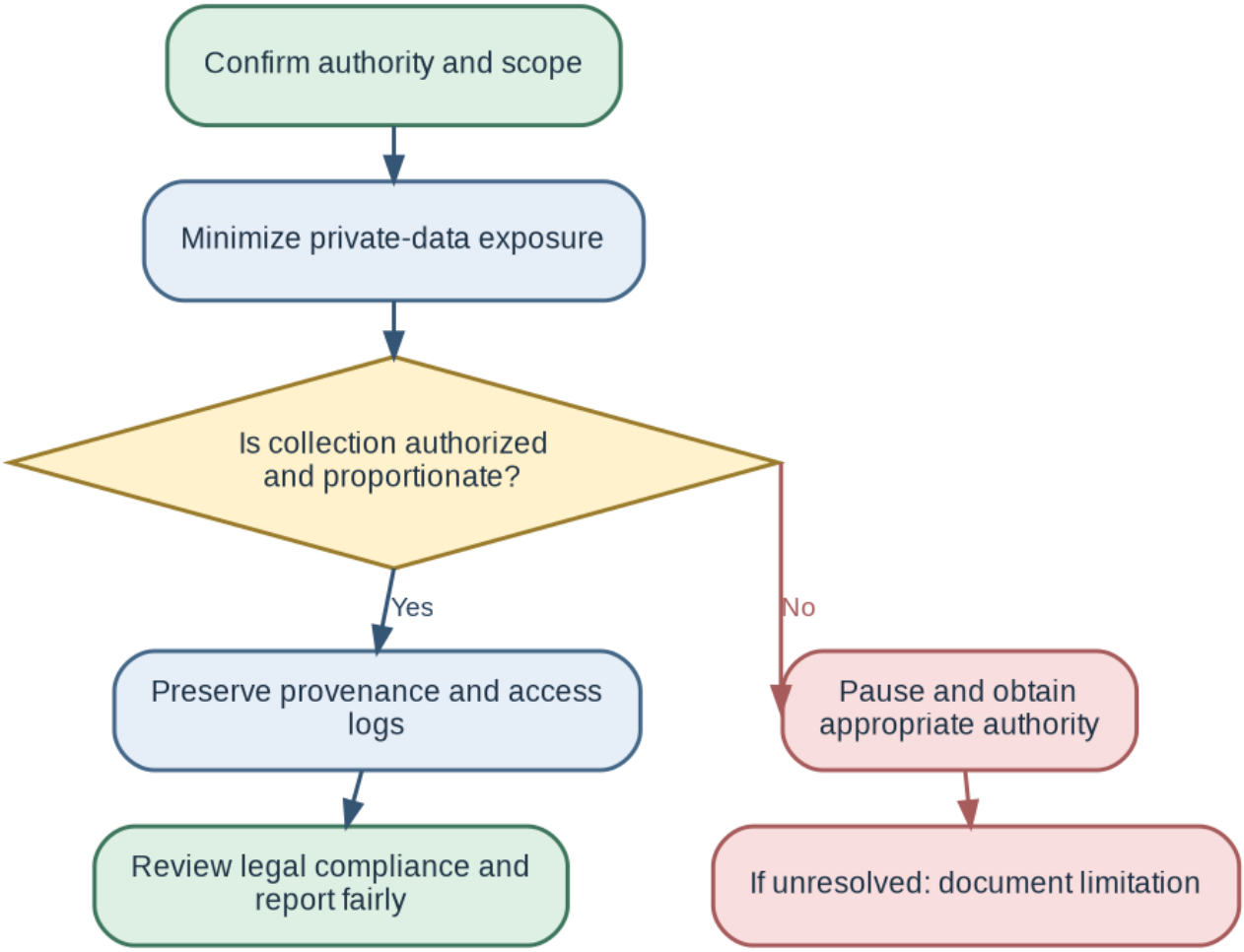
Section	Required contents
4. Findings	Each finding linked to exhibit ID, artifact path/offset, raw timestamp, interpretation, and corroboration.
5. Timeline and analysis	Normalized event chronology, assumptions, alternative hypotheses, and unresolved gaps.
6. Conclusion	What is supported, what is not supported, confidence, and limits of user attribution.
7. Appendices	Logs, hash manifests, relevant screenshots, export records, chain-of-custody forms, and technical details.

7

Briefly discuss the legal and ethical considerations involved in collecting digital evidence in a college environment.

The team should obtain institutionally valid authorization and, where required, consent or legal process before accessing accounts, devices, communications, or cloud records. Scope must be limited to the stated incident and relevant data. Shared staff computers and student records may contain unrelated personal or confidential information; access should be minimized, privileged or sensitive material handled under policy, and reports distributed only to authorized recipients. Investigators must avoid retaliation, conflicts of interest, and assumptions based on a person's status.

Preserve provenance, maintain accurate custody records, document methods, and retain the original data and examination notes. In India, evidentiary requirements for electronic or digital records should be assessed under applicable law, including the Bharatiya Sakshya Adhiniyam, 2023, in force from 1 July 2024; the exact requirements depend on the proceeding and evidence type [4]. Institutional legal counsel should determine applicable procedures. Ethical investigation means reporting exculpatory or contradictory artifacts as well as inculpatory material, protecting privacy, and not exceeding authority.



Flowchart 7. Question-specific reasoning sequence.

Assignment 2(2) - Linking a Suspicious File to a User Using File-System Evidence

Questions 8-14

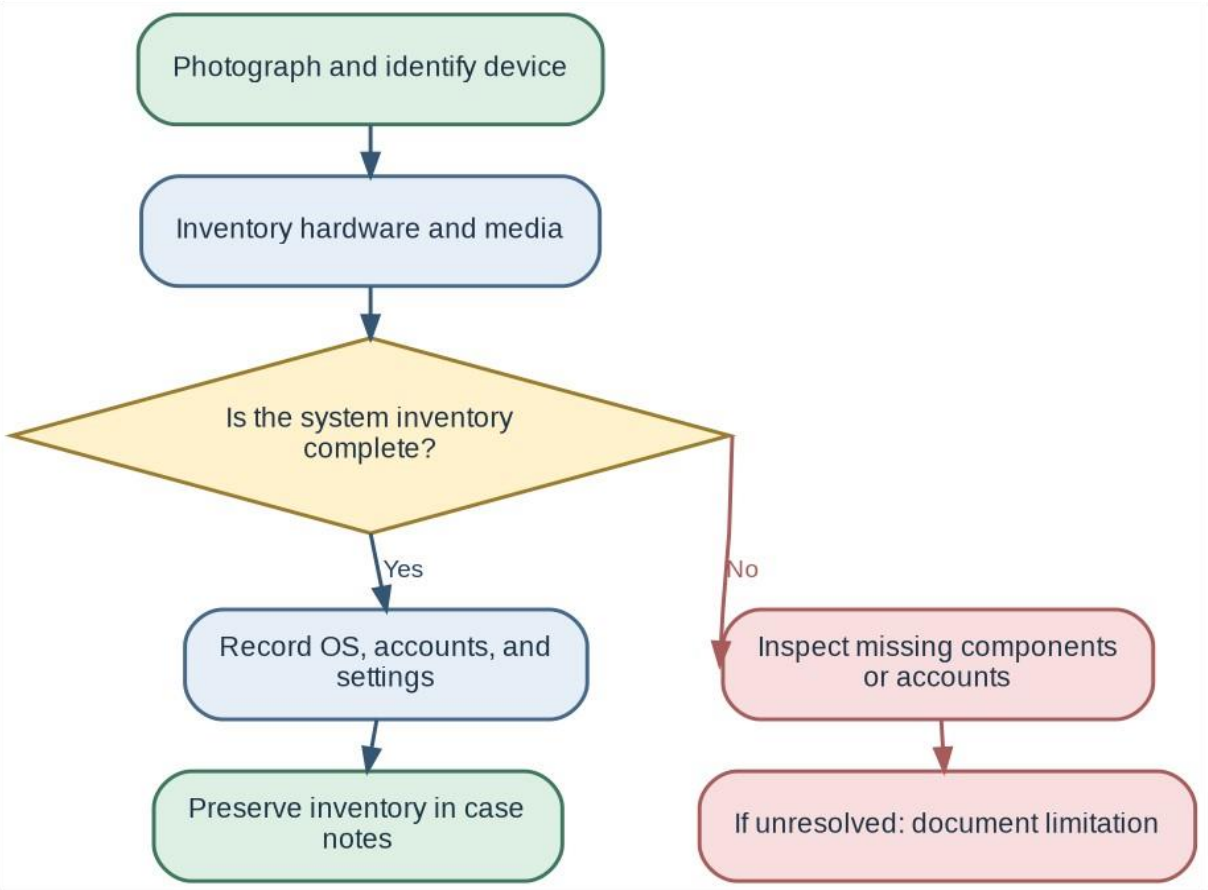
The shared desktop contains a suspicious executable, hidden files, encrypted documents, browser artifacts, and multiple user accounts. The goal is to test whether artifacts can link activity to a particular account while avoiding unsupported attribution to the person behind that account.

8

Identify the hardware and software components that should be documented before analysis.

The hardware inventory should identify the desktop manufacturer and model, asset tag, serial number, processor, installed RAM, internal storage type/capacity/serial number, attached USB or other removable devices, network interfaces, and any relevant peripherals. Photograph the physical setup and record the device's power state, connected cables, date/time shown, and visible damage. Record who supplied the device and any handling before seizure.

The software inventory should include operating-system edition, version, build, architecture, file system, disk encryption state, installed security products, relevant applications and browser versions, local and domain accounts, user-profile paths, system time zone, clock offset if known, and system configuration. Document forensic tool names, versions, validation status, acquisition format, and settings. Do not log in as a user or launch the suspicious executable to learn about it; use a forensic copy and controlled analysis environment.



Flowchart 8. Question-specific reasoning sequence.

Pre-analysis inventory checklist

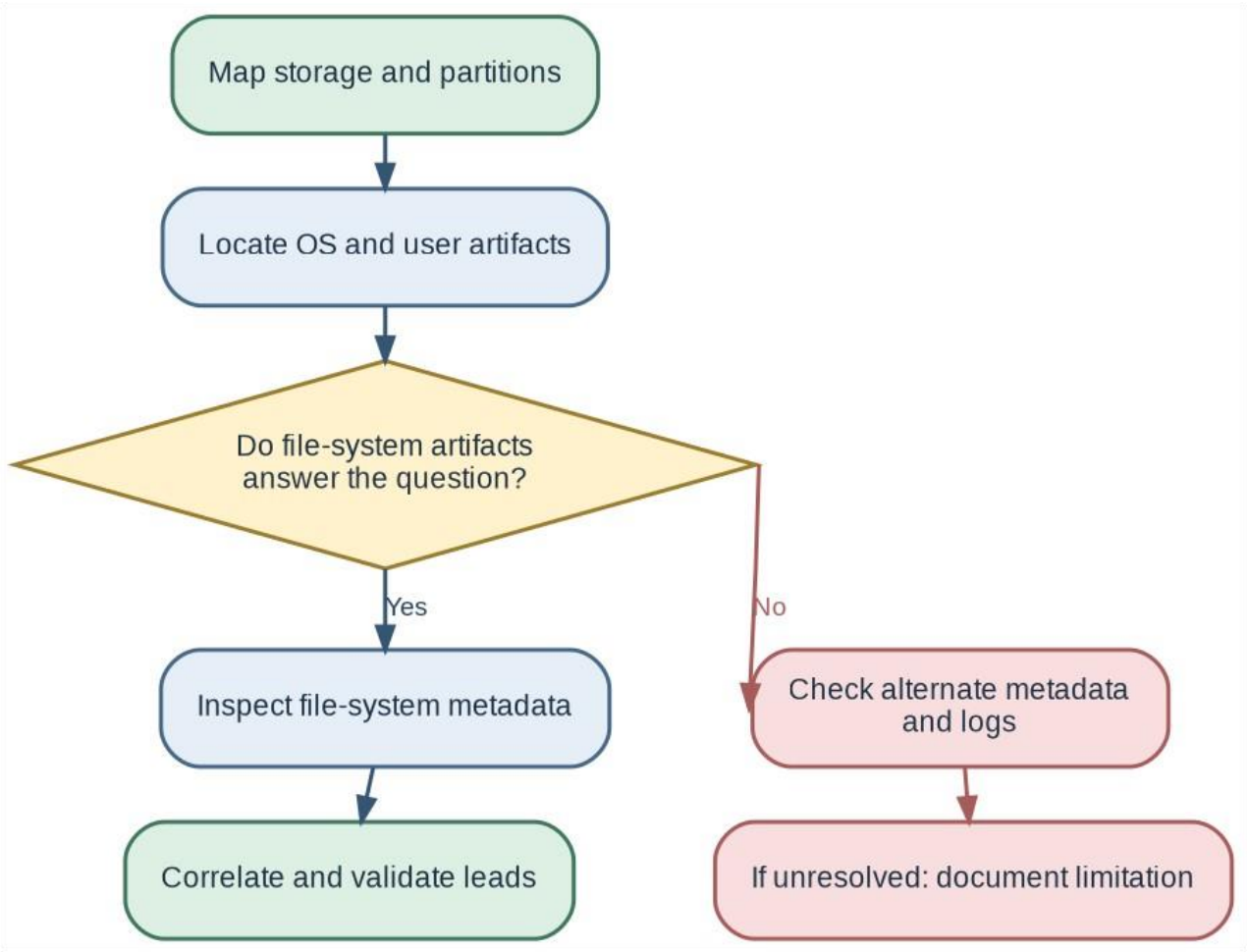
Category	Items to record
Hardware	Make/model, asset tag, serial, CPU/RAM, drive model/serial/capacity, interface, partitions, attached USB/media, network adapters.
Operating system	Name, edition, version/build, architecture, file system, patch level, time zone, clock offset, encryption state.
Accounts and applications	Local/domain accounts and SIDs, profile paths, browsers, suspicious program details, relevant security tools and versions.
Forensic environment	Acquisition and analysis tools/versions, validation status, write blocker, destination media identifiers, hash algorithm, case ID.

9

Explain how storage devices, operating systems, file systems, and metadata can help investigators locate relevant evidence.

Storage layout reveals where evidence may reside: internal drives, partitions, removable media, allocated files, unallocated space, slack space, and system or recovery partitions. The operating system supplies artifacts such as account records, event logs, registry entries, execution traces, mounted-device history, and application data. User-profile directories may separate activity by account but can be accessed by administrators or shared through applications.

File-system structures such as NTFS Master File Table entries, directory paths, the USN Change Journal, and link files may show file creation, renaming, movement, deletion, or access-related context. Metadata can include timestamps, size, owner/security identifiers, file attributes, alternate data streams, and cryptographic hashes.
Metadata is a source of leads, not an unquestionable record: time values can be altered, copied, rounded, or affected by application behavior. Correlate metadata with logs and independent artifacts.



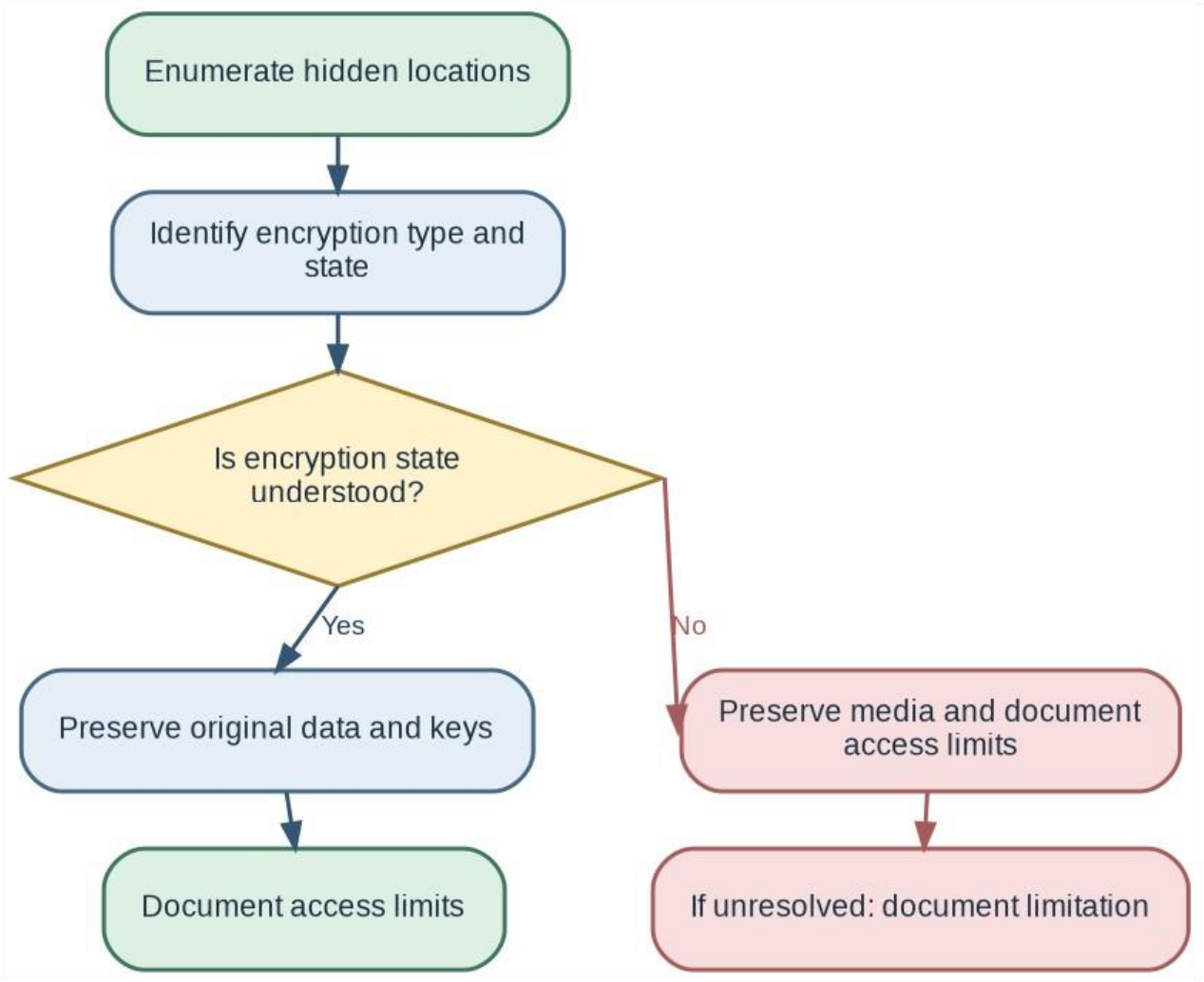
Flowchart 9. Question-specific reasoning sequence.

10

Describe how hidden files and encrypted files may affect a forensic investigation. Discuss appropriate evidence-handling considerations.

Hidden files may be excluded from ordinary directory views but remain discoverable through directory enumeration, file-system records, alternate data streams, application paths, or unallocated space. Their hidden attribute does not prove malicious intent. Encrypted files may reveal names, sizes, timestamps, access patterns, containers, or application references even when their contents cannot be read. Encryption may be full-disk, file-level, application-based, or cloud-side, and keys may exist only while a session is active.

Acquire and preserve data using authorized, forensically sound methods. Do not modify attributes, attempt repeated password guesses that may lock accounts, or bypass access controls without authority. Assess live-acquisition needs where a device is unlocked and decryption material is available, documenting any changes caused by that process. Preserve the original encrypted bytes and hash; seek authorized keys or passwords through lawful channels. Record which content was accessible, which was not, the tools and attempts used, and any uncertainty.



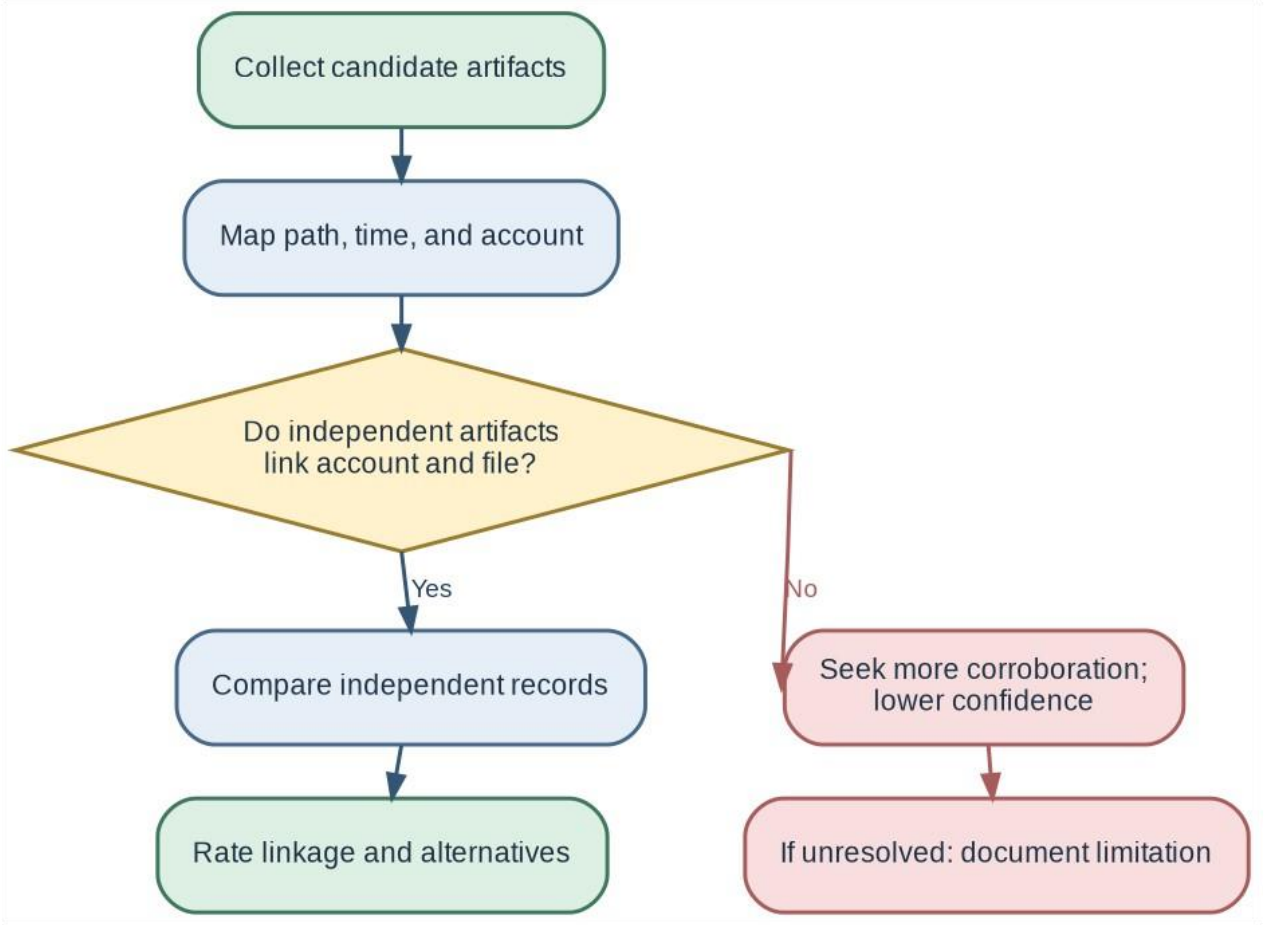
Flowchart 10. Question-specific reasoning sequence.

11

Prepare an evidence-linkage table showing how file names, paths, timestamps, user accounts, metadata, and other artifacts could be correlated to a user.

A linkage matrix should pair each artifact with the proposition it may support, the account or device context, and its alternative explanations. Strong linkage generally comes from several independent sources agreeing on the same user session, file, operation, and time; a single path or filename is rarely enough. Correlate account SIDs, logon sessions, file paths, execution artifacts, USB history, browser download records, and server-side events where available.

Separate association from attribution. A file in a user's profile suggests that the profile had access to it, not necessarily that the account holder authored or opened it. An execution trace may establish that a program ran on the system but may not establish which person was at the keyboard. State the confidence level and identify what evidence could strengthen or weaken each link.



Flowchart 11. Question-specific reasoning sequence.

Evidence-linkage matrix

Artifact / source	Possible relationship to account	Strength and limitation
Security logon record with account SID and session ID	Can associate activity with a Windows account/session.	Strong for account session if log integrity is established; not proof of who sat at keyboard.
NTFS MFT entry / USN Change Journal	May show path, record sequence, file changes, rename, or deletion.	Supports file-system activity; timestamps and user attribution are limited.
Prefetch / Amcache or equivalent execution artifact	May indicate a program was run or inventoried on the system.	Execution context varies by artifact and OS version; may not identify operator.
LNK file / Jump List	May record a file path accessed through a shell/application context.	Corroborative; generated/updated by application behavior and can be copied.

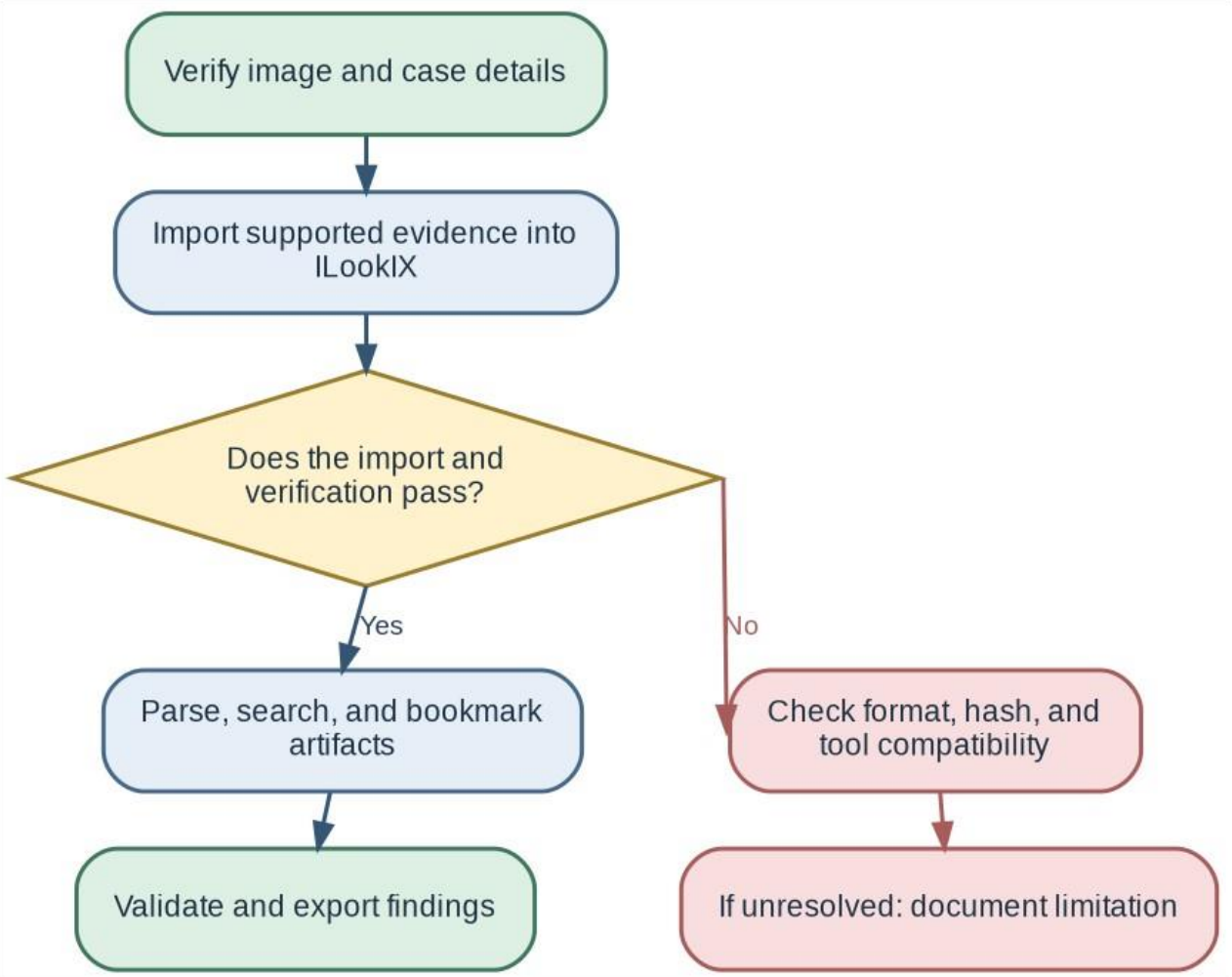
Artifact / source	Possible relationship to account	Strength and limitation
Browser download record plus matching file hash	May connect a file to a URL and browser profile.	Useful linkage; a profile may be shared or synchronized.
USB connection history plus matching file on USB	May link removable-media use to the host and support a copy hypothesis.	Connection does not prove a particular file transfer without additional evidence.
Server or cloud audit record	May record authenticated access to a specific resource.	Potentially direct for an account action; credential compromise remains possible.

12

Explain a suitable workflow for importing/acquiring the forensic evidence and analyzing it using ILookIX, based on the concepts covered in the syllabus.

First verify the acquisition record, image/container format, source identifiers, hash values, and tool version; create a case using a unique identifier and examiner details. Import the verified forensic image or supported evidence container into ILookIX following the version-specific user guide. If the interface supports verification at import, compare the reported values with acquisition notes and record any mismatch before continuing.

Set the correct time zone and parsing options, then enumerate partitions, file systems, user profiles, hidden files, deleted entries, metadata, and relevant artifacts. Search for known filenames, hashes, paths, and keywords; examine suspicious executables in a controlled workflow rather than running them on the evidence host. Bookmark relevant results, export reports with source paths and offsets where available, preserve tool logs, and independently validate key findings. Menu names and supported formats vary by release, so the precise clicks must follow the installed version's documentation; this workflow does not claim that ILookIX was actually run.



Flowchart 12. Question-specific reasoning sequence.

LookIX workflow record

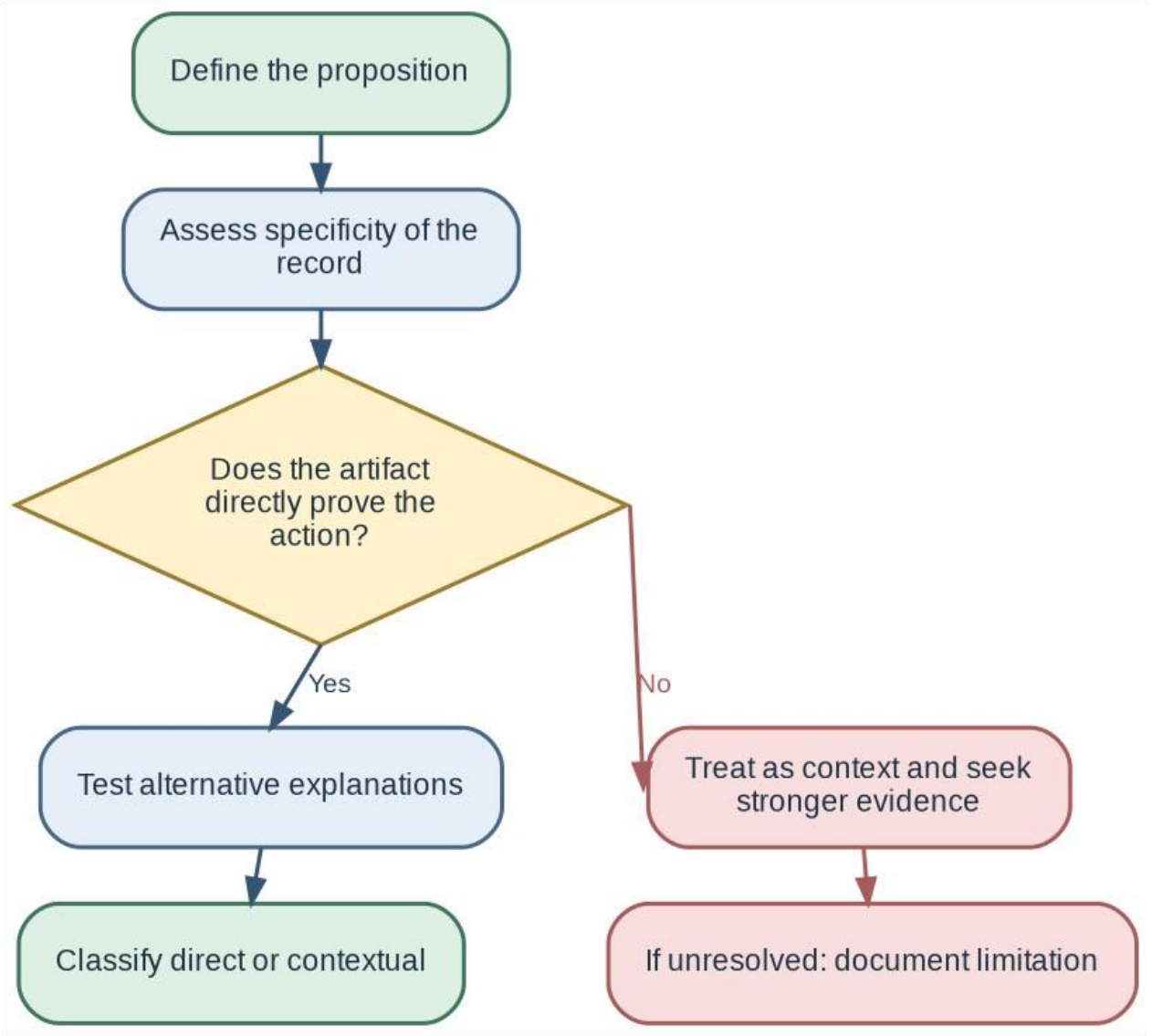
Stage	Action	Record to preserve
1. Validate input	Confirm image/container type, source ID, hash, acquisition notes, and known errors.	Input manifest and verification result.
2. Create case/import	Create a unique case and import only a supported verified evidence copy.	Case settings, tool/version, import log.
3. Configure and parse	Set time zone and supported parsing options; enumerate file systems and user profiles.	Settings, parse log, errors.
4. Examine and search	Search names, hashes, paths, keywords, hidden/deleted artifacts, and relevant metadata.	Queries, bookmarks, source paths/offsets.
5. Validate and report	Cross-check key artifacts, export results, and compare findings to independent evidence.	Report, exports, hashes, limitations.

13

Differentiate between an artifact that directly links a user to an activity and an artifact that merely provides supporting/contextual information.

An artifact that directly records an authenticated operation is more probative than a general clue. For example, a server audit entry may record that account A read a specific examination file at 22:14 from session ID S, and an independent authentication log may associate that session with a workstation. It directly links the operation to an account/session, although it still may not identify the human operator if credentials were shared or compromised.

Contextual artifacts include a file located in a user's Downloads folder, a common browser history entry, a filename that resembles a user's project, or a USB device previously connected to the computer. These artifacts help explain the environment but admit alternative explanations such as shared use, automated processes, syncing, or copying by another user. Classify each artifact as direct, corroborative, or contextual in relation to a clearly stated proposition, and avoid treating categories as absolute.



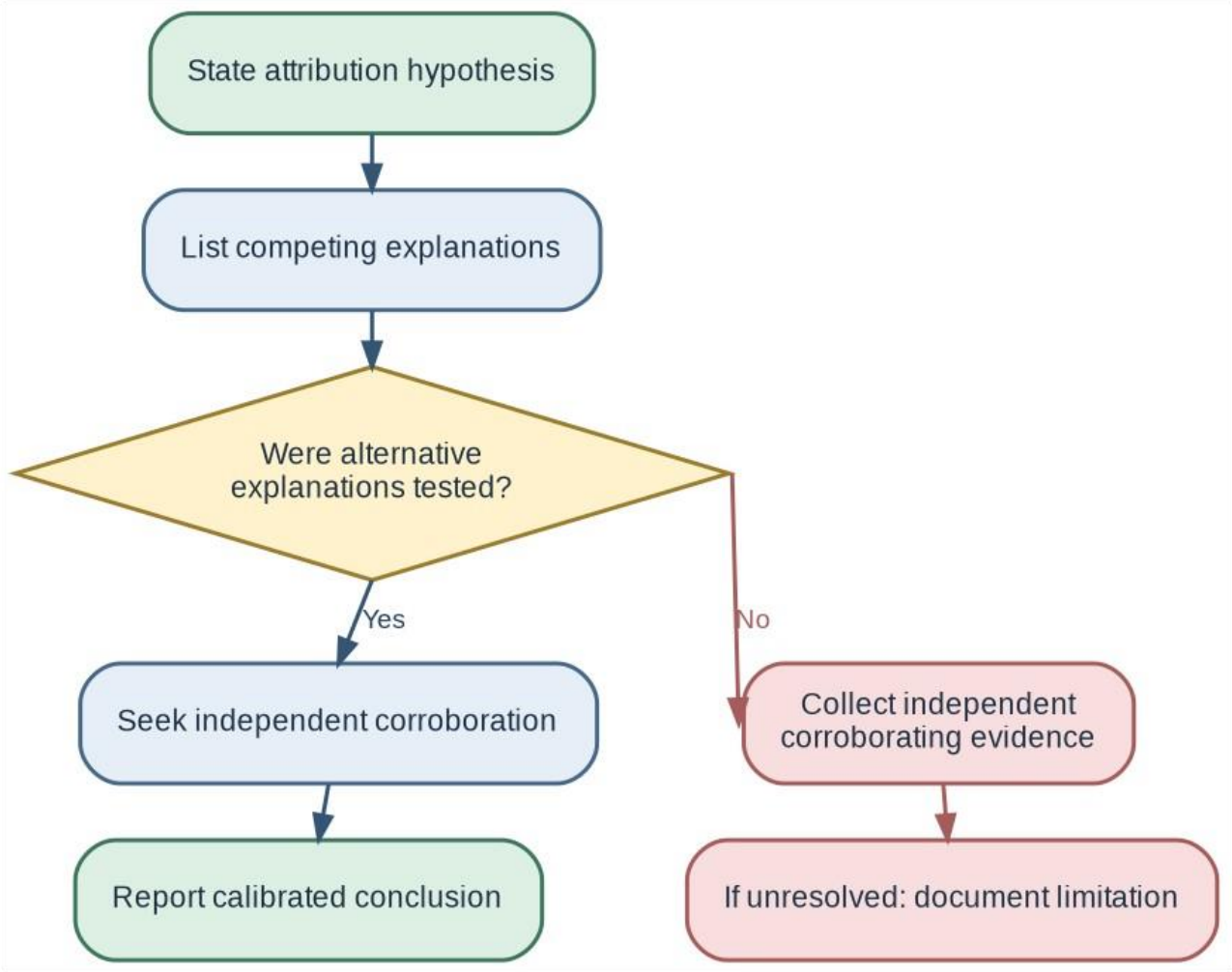
Flowchart 13. Question-specific reasoning sequence.

14

Identify at least three limitations or alternative explanations that investigators should consider before attributing the activity to a particular user.

First, the desktop may be shared and another person may have used an unlocked session. Second, the account may have been compromised through stolen credentials, malware, remote access, or an unattended logged-in session. Third, timestamps may reflect a wrong clock, time-zone conversion, copying, or application-specific semantics. Fourth, artifacts may be created automatically by background services, synchronization, or security software rather than by deliberate user action.

Other limitations include incomplete or rotated logs, a file copied from another system, a portable application running without installation, a false-positive malware classification, or tool parsing errors. Investigators should corroborate each material claim with independent evidence, document gaps, test both inculpatory and exculpatory hypotheses, and express conclusions proportionately. The defensible conclusion may be that activity is associated with an account or device while the identity of the individual remains unresolved.



Flowchart 14. Question-specific reasoning sequence.

Assignment 2(3) - Forensic Imaging and Safekeeping of a Suspect Storage Device

Questions 15-22

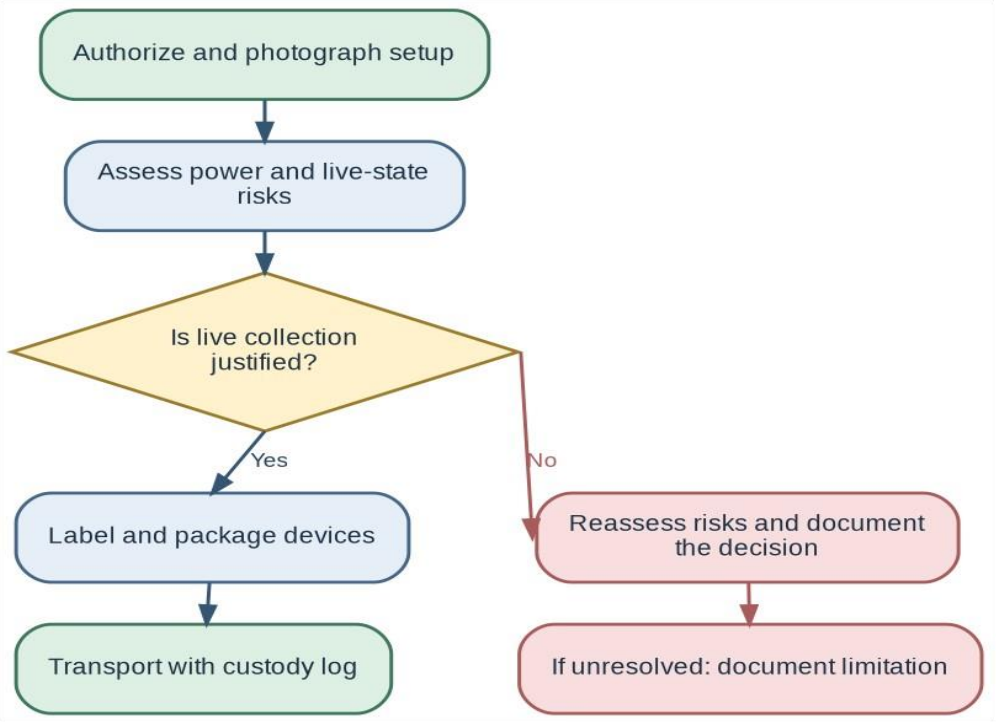
A seized laptop contains an internal storage device and an attached USB drive. The investigation requires careful physical collection, imaging, integrity verification, possible file carving, and documented custody. IXImager and the ASB container are considered in the supplied case study.

Question 15 15

Prepare a step-by-step procedure for safely collecting and transporting the laptop and USB device before imaging.

Obtain authority and assign evidence identifiers. Photograph the laptop, screen, ports, connected USB drive, cables, network connections, and visible power state before touching anything. Record make/model, serials, date/time, location, and the person who collected each item. If the laptop is on, do not reflexively power it down: a trained examiner should assess whether volatile data or live encryption keys require justified live acquisition, and document any live-response actions.

If the device is powered off, keep it off. Disconnect or isolate network connectivity when safe and authorized; avoid remotely wiping or syncing devices. Remove the USB only after documenting its insertion and following the scene plan. Package each item separately in suitable protective packaging, label all seals, and protect from impact, static, moisture, and unauthorized access. Transport securely and log every transfer, seal number, time, sender, receiver, and purpose. Do not place a laptop and loose USB together in a way that can cause loss or damage.



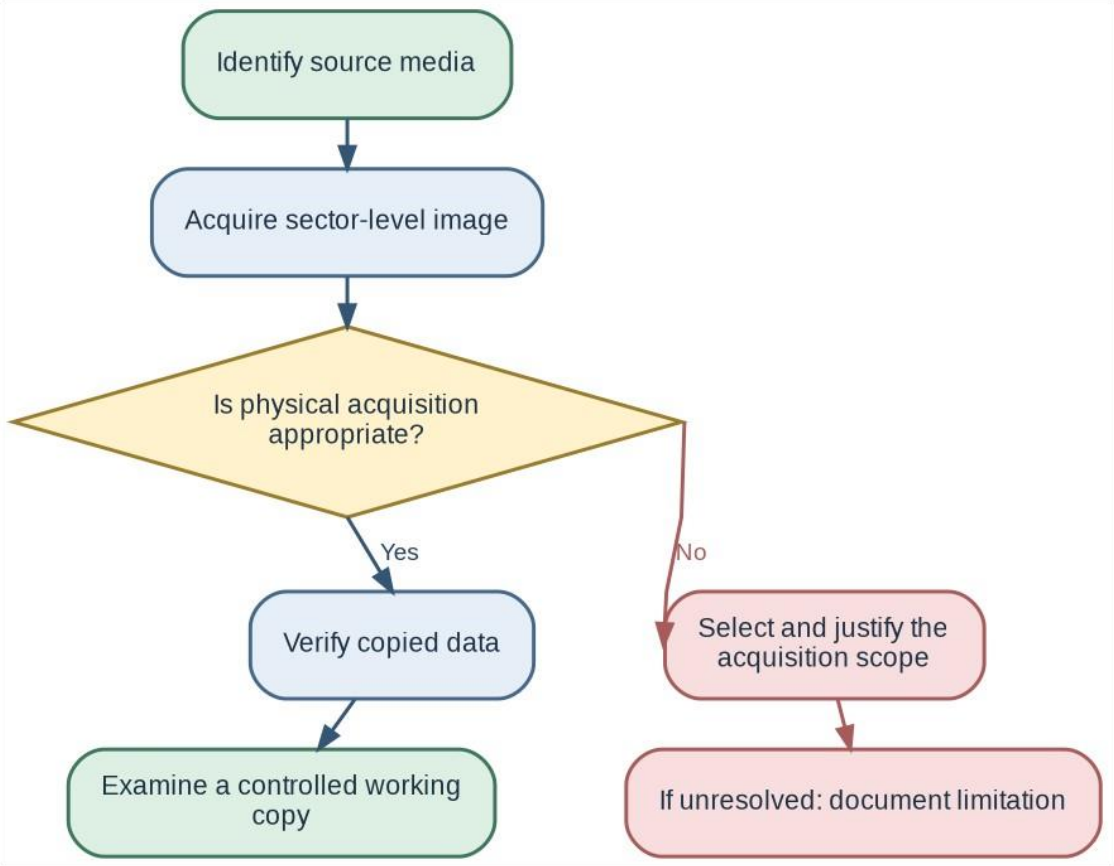
Flowchart 15. Question-specific reasoning sequence.

16

Explain the purpose of physical acquisition and why analysis should normally be performed on a forensic image rather than the original storage media.

Physical acquisition attempts to preserve a sector-level copy of the accessible source media, including allocated files, deleted-file remnants, unallocated space, file-system structures, and slack space. This gives investigators a broader basis for recovery than copying only visible files. Acquisition must document unreadable sectors, encryption, damaged media, HPA/DCO or other device-specific limits where relevant, and whether the copy is a full physical image or a logical collection.

Analysis is normally conducted on a verified forensic image because it reduces writes to the original, allows repeatable examination, and permits multiple examiners to work from controlled copies. The original should be sealed and stored safely. Hashes and acquisition logs help show that the working copy corresponds to the acquired data. A bit-for-bit image can still contain pre-existing changes, unreadable sectors, or data outside its scope; imaging does not by itself prove who created the data or make the evidence legally admissible.



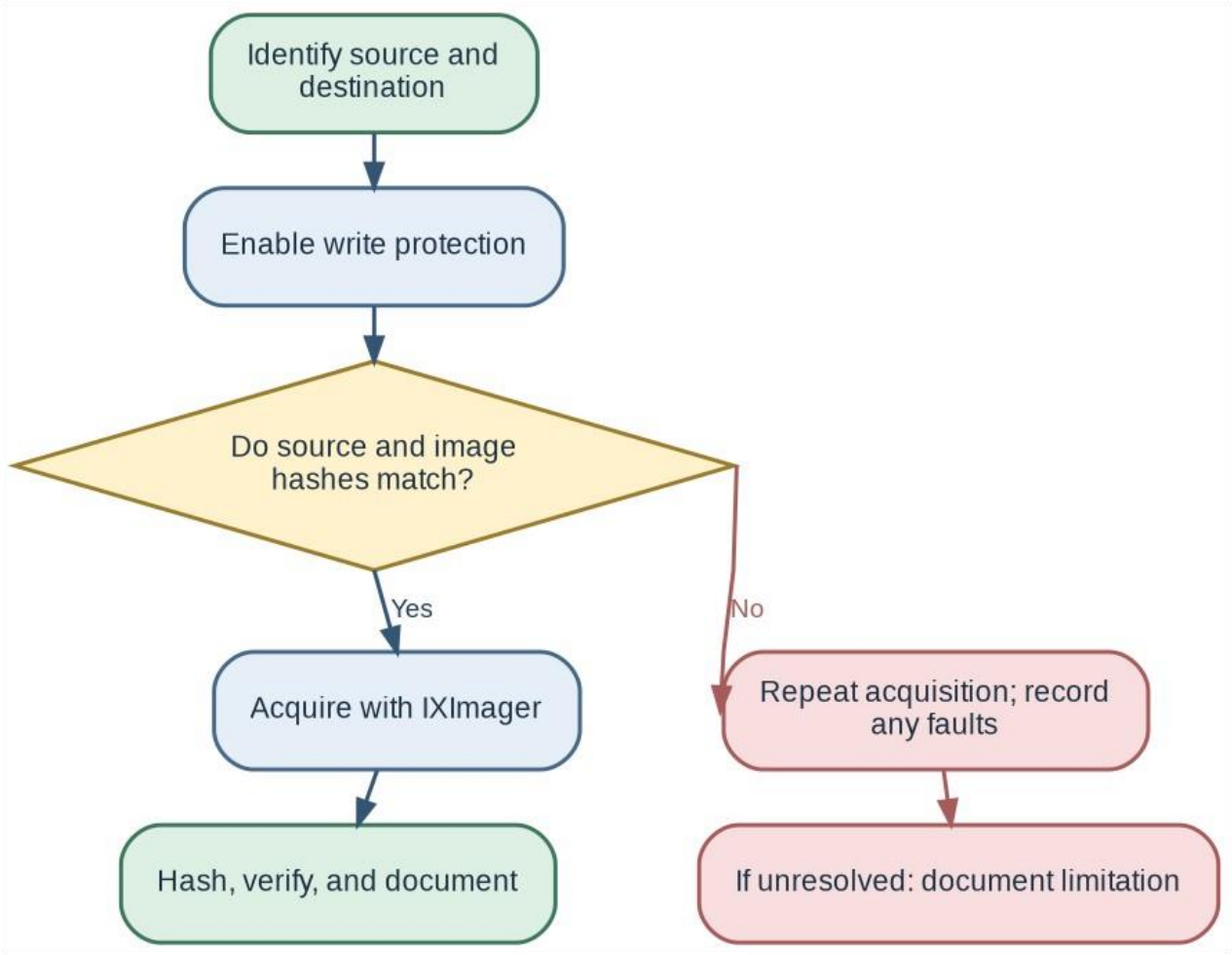
Flowchart 16. Question-specific reasoning sequence.

17

Design a forensic imaging workflow using IXImager. Include source identification, write protection, imaging, verification, hashing, and documentation.

Start by documenting the source drive's make, model, serial number, capacity, interface, logical sector size, and physical condition. Validate the IXImager version and destination format for the task. Attach the source through a compatible hardware write blocker and confirm that the source is not writable; select a separate destination with enough capacity and identify it by serial number to avoid reversing source and destination.

Configure acquisition options and a case identifier, then image the source to a supported destination or ASB evidence container using the documented IXImager procedure. Record progress, read errors, retries, tool version, operator, start/end time, and any deviations. Calculate the required cryptographic hashes of source data and acquired image according to the tool's supported verification method. Re-read/verify the image where supported and compare results. A container-file hash and a hash of the represented source data may differ because they cover different byte sequences; label each clearly. Seal the original and preserve the log with the image.



Flowchart 17. Question-specific reasoning sequence.

IXImager acquisition checklist

Stage	Action	Verification record
Source identification	Record source model/serial/capacity/interface/sector size and physical condition.	Photographs and source inventory.
Write protection	Connect via compatible hardware write blocker and confirm write protection.	Blocker model/serial, indicator/test.
Destination and settings	Confirm destination identity/capacity and choose supported image/ASB format.	Destination media ID, settings, case ID.

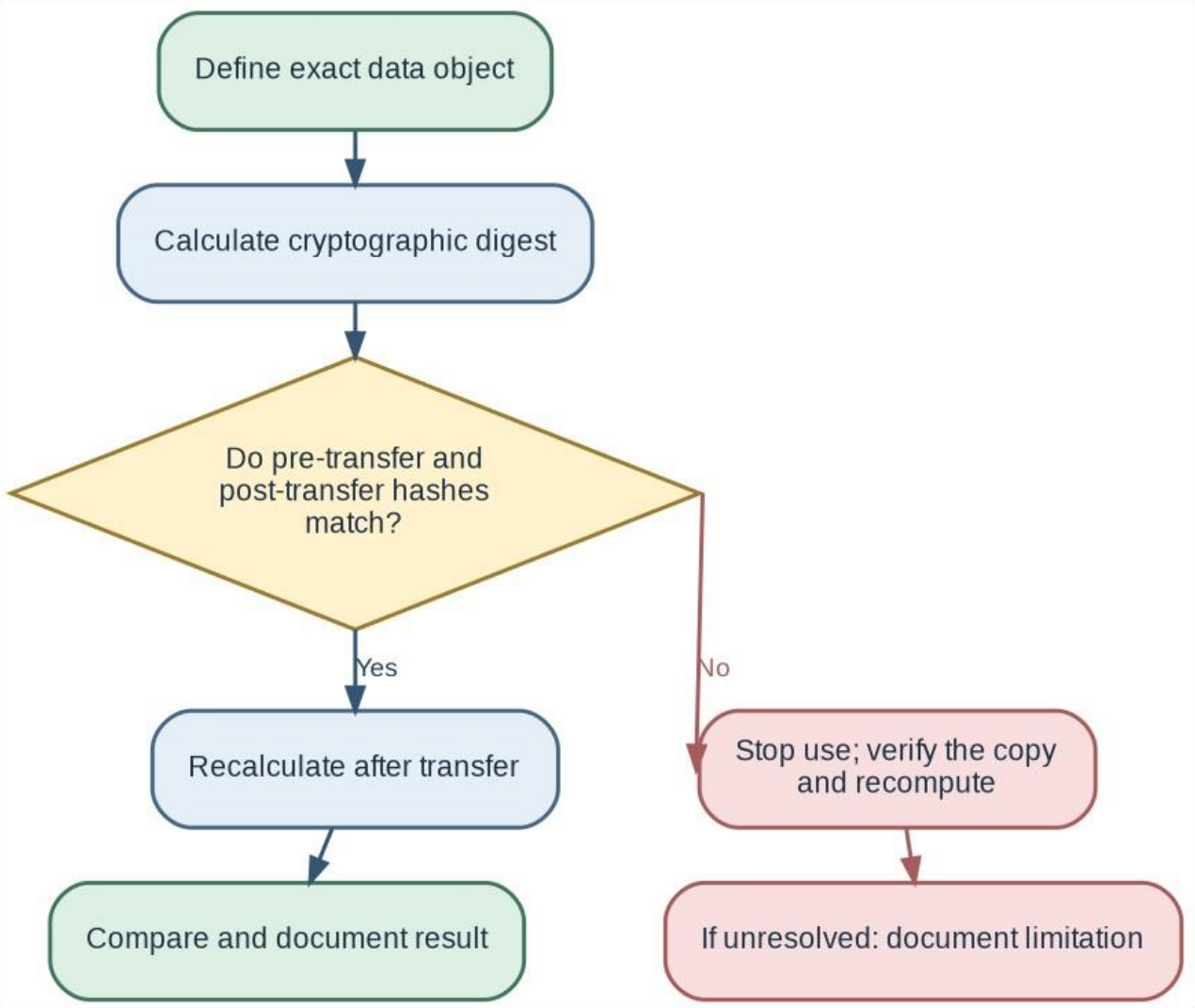
Stage	Action	Verification record
Acquisition	Run the version-specific IXImager procedure; monitor errors and retries.	Tool/version, operator, start/end time, acquisition log.
Hash and verification	Calculate and compare consistent data hashes; verify container and payload as applicable.	Algorithm, object hashed, values, verification outcome.
Safekeeping	Seal original, store image securely, and preserve a separate manifest.	Custody entries, seal IDs, storage path.

18

Explain how hash values can be used to demonstrate integrity before and after acquisition.

A cryptographic hash such as SHA-256 produces a digest for a defined byte sequence. If the source data and the corresponding acquired data are hashed consistently and the values match, this supports the conclusion that the compared data are identical at the time of hashing. The report must identify the algorithm, exact object hashed, acquisition tool, and stage of calculation. Record the original acquisition log, verify the image after transfer, and recalculate hashes of working copies when moved or exported.

Hashing is not a substitute for custody records or tool validation. A matching hash does not prove the data were truthful before collection, that the device clock was correct, or that the data were created by a particular user. For a compressed or structured ASB container, the whole-container hash may not match the source-media digest; the report should distinguish the container hash from any hash of the image payload or source sectors. If values differ, stop reliance on the copy, preserve both versions, investigate the discrepancy, and document it.



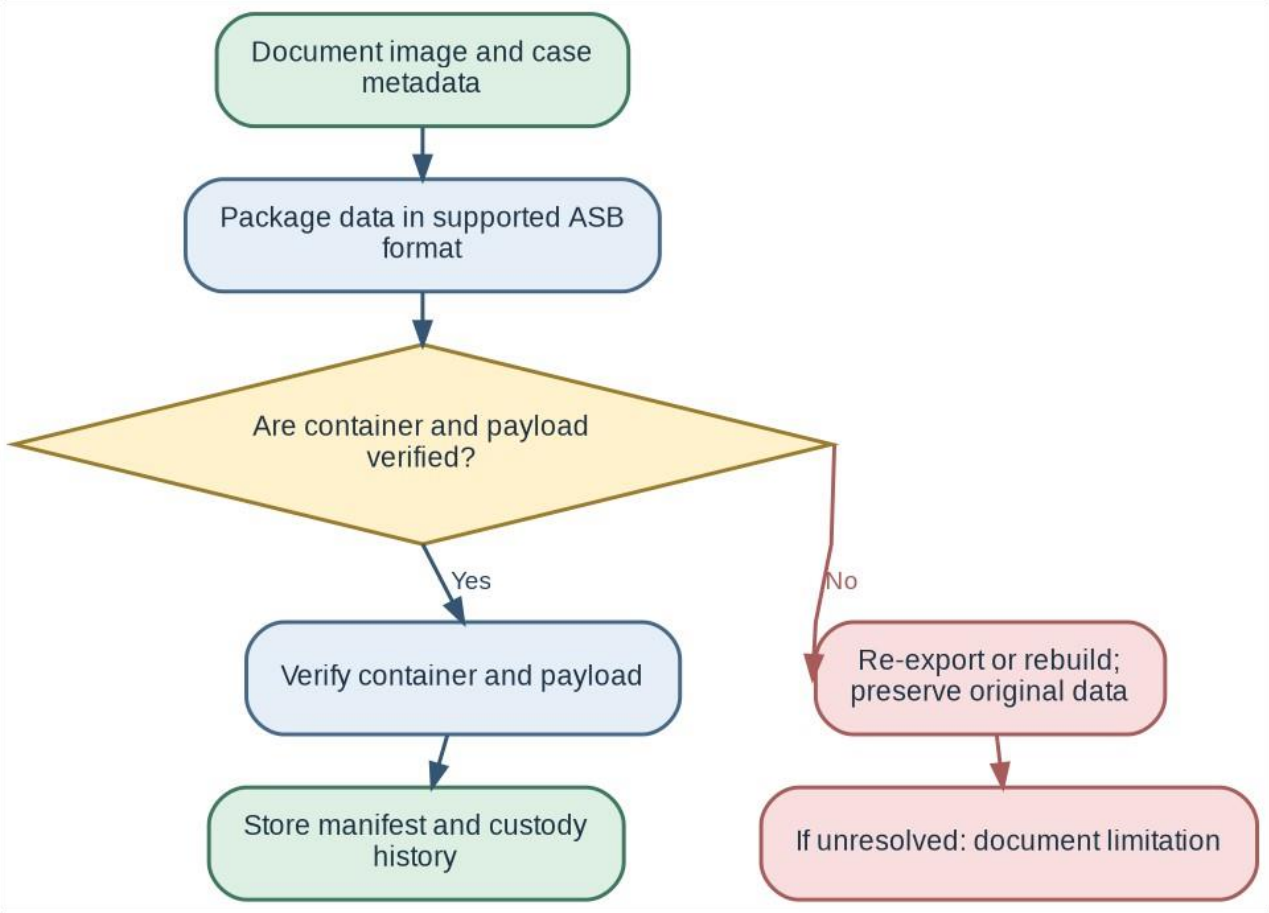
Flowchart 18. Question-specific reasoning sequence.

19

Describe the role of an ASB container in storing or managing forensic evidence and identify the information that should accompany the evidence.

In the IXImager/ILookIX workflow described in forensic training material, an ASB container is a tool-associated evidence format used to package acquired data with acquisition information and supporting integrity records. The exact structure and metadata depend on the tool version, so the examiner should confirm supported features using the installed version's documentation rather than treating ASB as a universal interchange standard. Container packaging can improve evidence management, but it does not replace an independently maintained chain of custody.

Accompany the container with the case number, evidence identifier, source device details, acquisition type and scope, source/destination identities, tool and version, operator, date/time and time zone, write-blocker details, hash algorithm and values, verification results, read errors, acquisition settings, and custody history. Preserve a separate manifest and export logs in a secure location. Test the container's integrity and readability on a controlled copy. Clearly distinguish the hash of the container file from hashes representing the acquired source data.



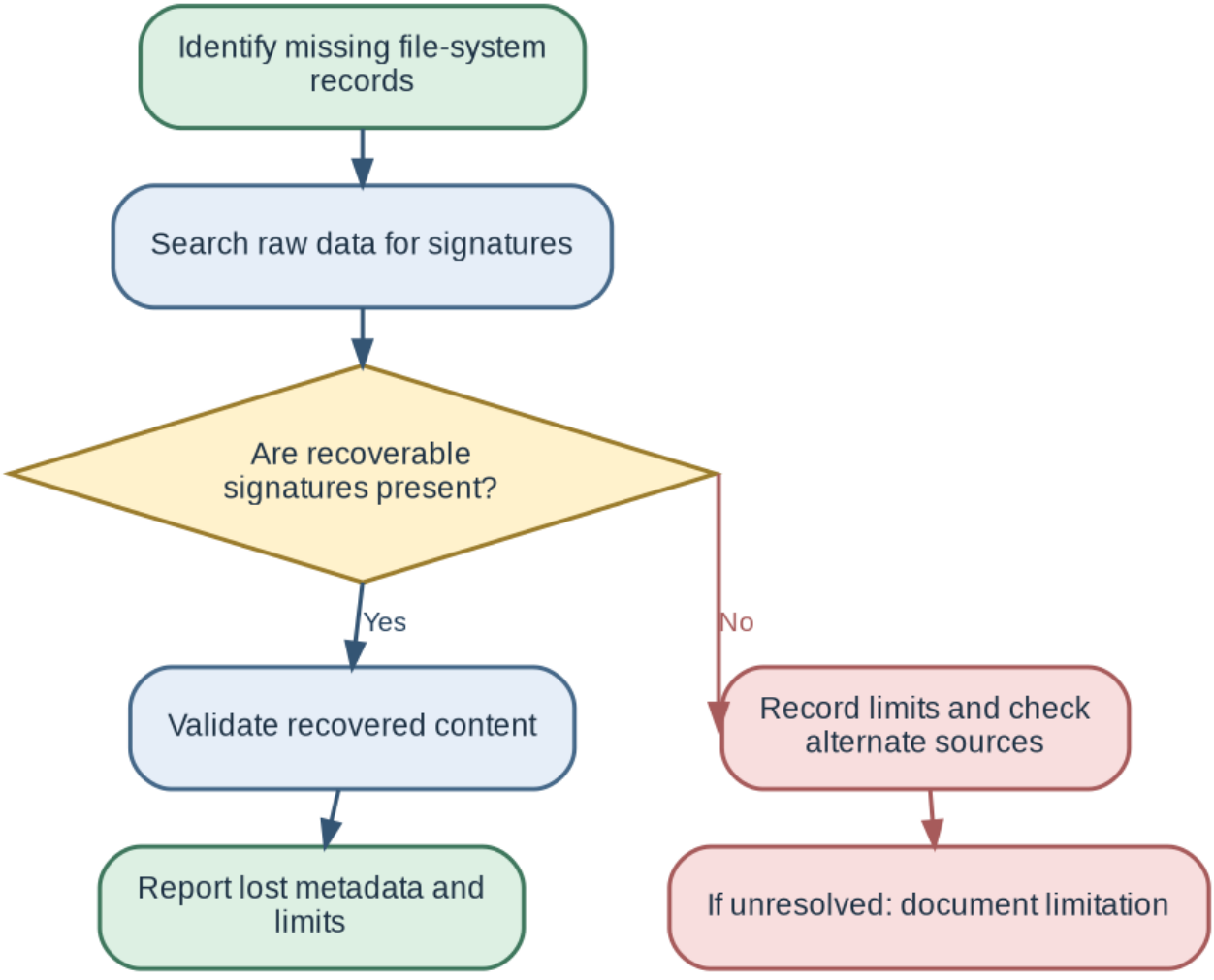
Flowchart 19. Question-specific reasoning sequence.

20

Explain data carving and discuss when carving may be necessary if file-system information is unavailable or damaged.

Data carving searches raw data for file signatures, structures, or content patterns to recover files without relying fully on directory entries or file-system metadata. It is useful when directory records are deleted, a file system is corrupted, partitions are missing, or unallocated space may contain remnants. Signature-based carving may recover file content even when the original filename, path, timestamps, and ownership metadata are no longer available.

Carving can produce incomplete files, false positives, fragments, or files assembled from unrelated data. Fragmented files are difficult to reconstruct reliably using simple signatures. Investigators should carve from a verified copy, record tool/version and carving parameters, preserve offsets and source regions, hash recovered outputs, and validate content with file-format parsers or independent context. A carved image or document should not be attributed to a user without corroborating records. If file-system metadata exists, retain it and use carving as a complementary technique rather than discarding context.



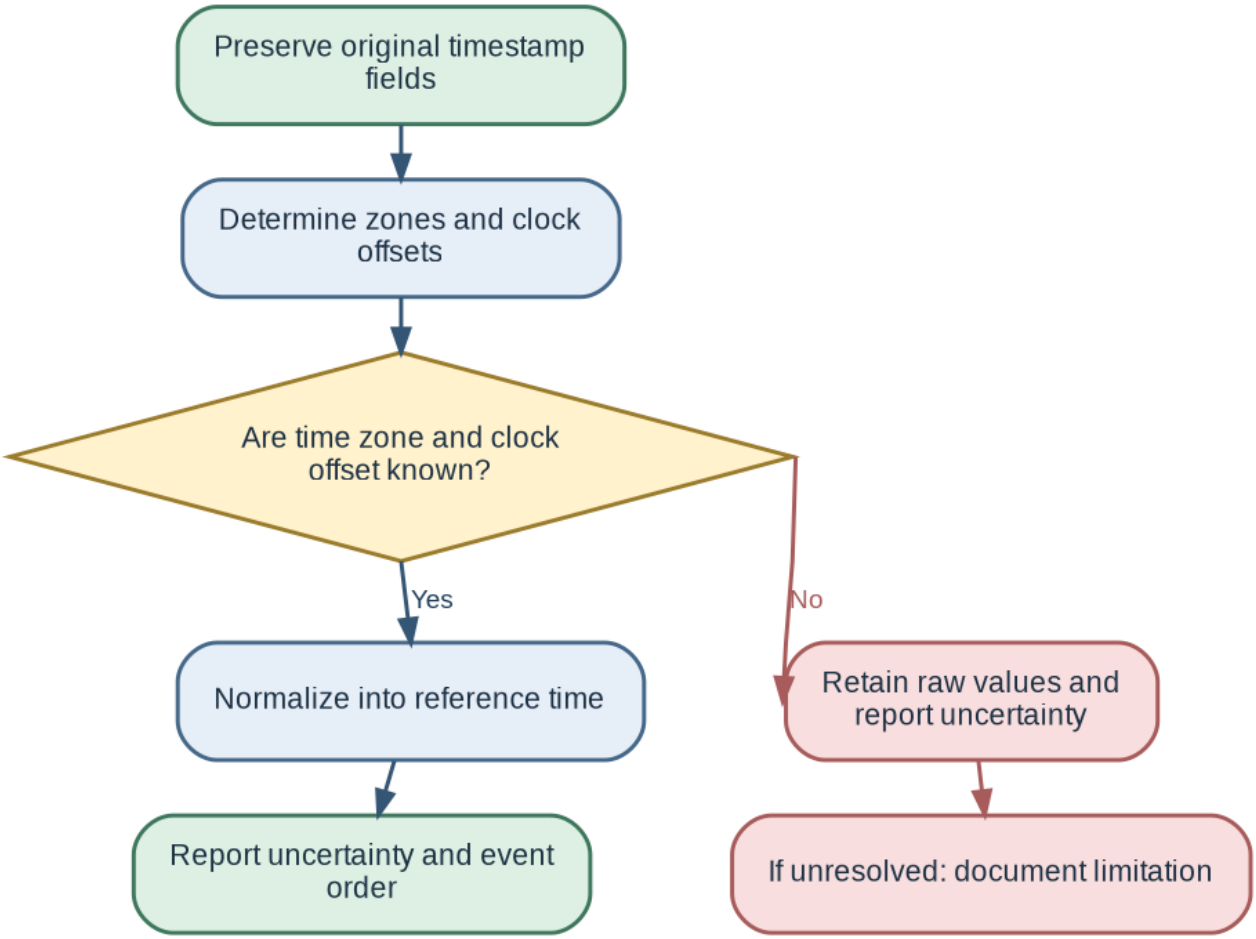
Flowchart 20. Question-specific reasoning sequence.

21

Discuss date/time problems in digital evidence and explain how investigators should document time-zone, clock, and timestamp uncertainties.

Digital artifacts may record local time, UTC, epoch values, file-system-specific time values, or server-generated times. Systems can have incorrect clocks, drift, manual adjustments, daylight-saving changes, or differing time zones. Copying and synchronizing files can change some timestamps, while access times may be disabled or updated inconsistently. Logs from cloud services may use server time and display it in the viewer's local zone.

Record the raw timestamp, its source field and format, the system's configured time zone, daylight-saving offset, known clock skew, collection time, and conversion method. Normalize a separate analysis column to UTC or another stated reference standard; never overwrite the raw value. Use trusted events to estimate skew and express uncertain ordering when two events fall within the uncertainty interval. Report that timestamps describe recorded system events, not necessarily the time a person consciously acted.



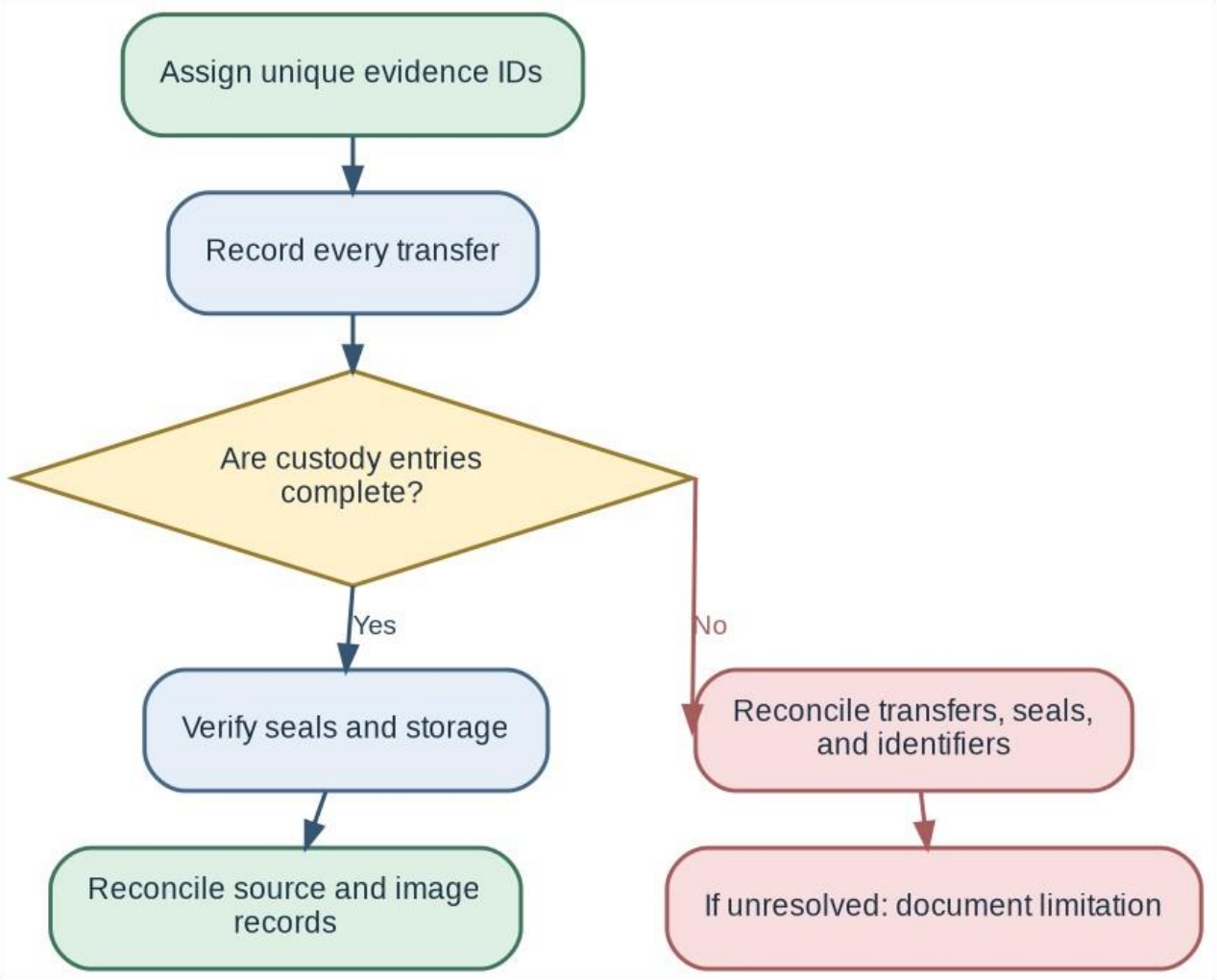
Flowchart 21. Question-specific reasoning sequence.

22

Prepare a chain-of-custody table for the seized laptop from collection through laboratory examination.

A chain-of-custody form provides a chronological record of who controlled an item, when control changed, where it was stored, why it was transferred, and whether seals remained intact. Use a unique identifier for the laptop, its internal drive, and its attached USB device; do not treat them as one item without separate labeling. Every participant should sign or authenticate the transfer entry, and corrections should preserve the original entry.

The table below is a blank model to complete with actual collection details. Never invent the time, signature, seal condition, or transfer recipient. Include the evidence ID on packaging and in acquisition logs. When a forensic image is created, assign it its own identifier and record its hash and storage location, linking it to the source evidence ID. Unexplained custody gaps must be investigated and disclosed.



Flowchart 22. Question-specific reasoning sequence.

Chain-of-custody form (blank model)

Date/time + zone	From / to	Purpose / location	Seal condition	Signatures / initials
[Enter actual value]	[Collector] -> [Evidence custodian]	Collection at scene; item ID: LAP-01	[Record actual state]	[Signatures]
[Enter actual value]	[Custodian] -> [Transport officer]	Secure transport to laboratory	[Record actual state]	[Signatures]
[Enter actual value]	[Transport officer] -> [Lab receiver]	Laboratory receipt and secure storage	[Record actual state]	[Signatures]
[Enter actual value]	[Lab receiver] -> [Examiner]	Authorized imaging / examination	[Record actual state]	[Signatures]

Date/time + zone	From / to	Purpose / location	Seal condition	Signatures / initials
[Enter actual value]	[Examiner] -> [Evidence custodian]	Return of original; image assigned ID IMG-01	[Record actual state]	[Signatures]

Assignment 2(4) - Recovery of Deleted Evidence from Volume Shadow Copies

Questions 23-30

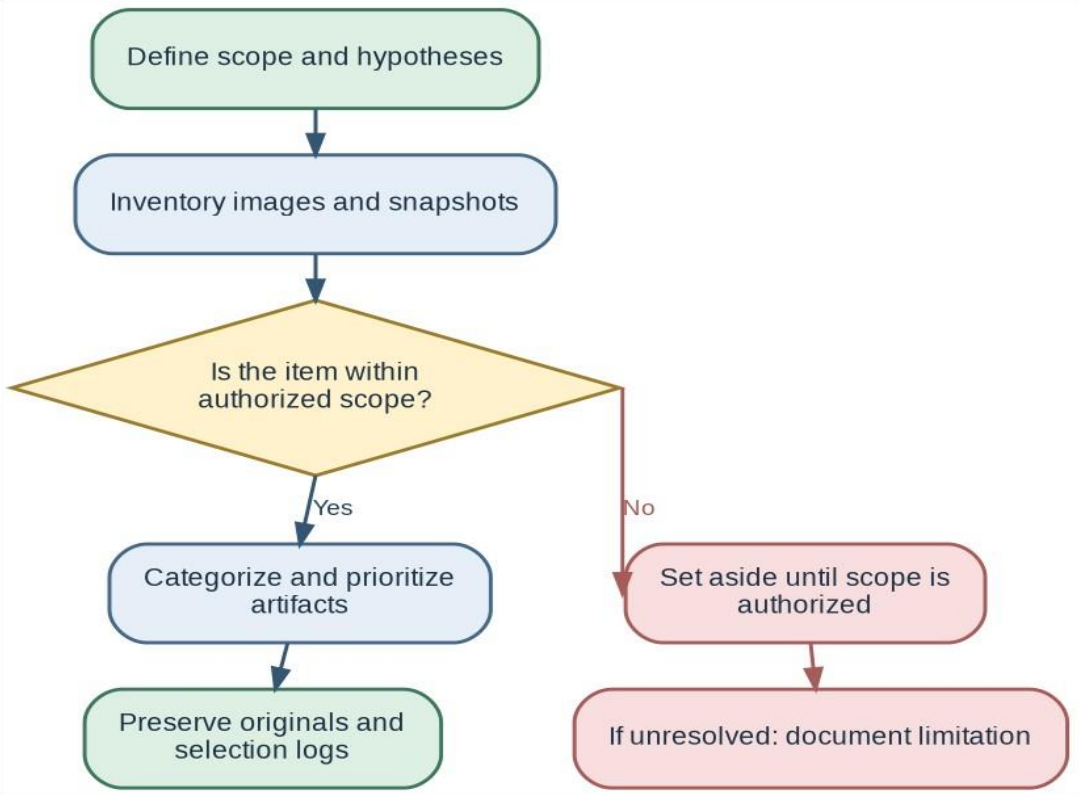
A forensic image of a Windows system is available. Volume Shadow Copies may contain previous versions of documents or artifacts deleted or modified after the snapshot was created. Recovery, interpretation, and validation should be done from a verified working copy, with uncertainty documented.

23

Explain how investigators should select and categorize digital evidence before detailed analysis.

Begin with the incident hypothesis and the scope authorized for examination. Identify relevant dates, users, file types, applications, paths, and systems, then inventory the full forensic image and available snapshots. Categorize artifacts by source, relevance, evidentiary role, and status (current, deleted, recovered, or contextual). Use hashes, filenames, paths, metadata, and known indicators to prioritize likely evidence without excluding unexplained or contradictory items prematurely.

Filtering should reduce noise in the analysis view, not destroy or permanently delete data from the original image. Preserve the acquisition and full image, document search terms and filters, and retain enough provenance to reproduce the selection. Mark irrelevant or out-of-scope data and protect it from unnecessary exposure.
Investigators should revisit categories when a lead changes the scope or reveals that a previously low-priority artifact is connected to the incident.



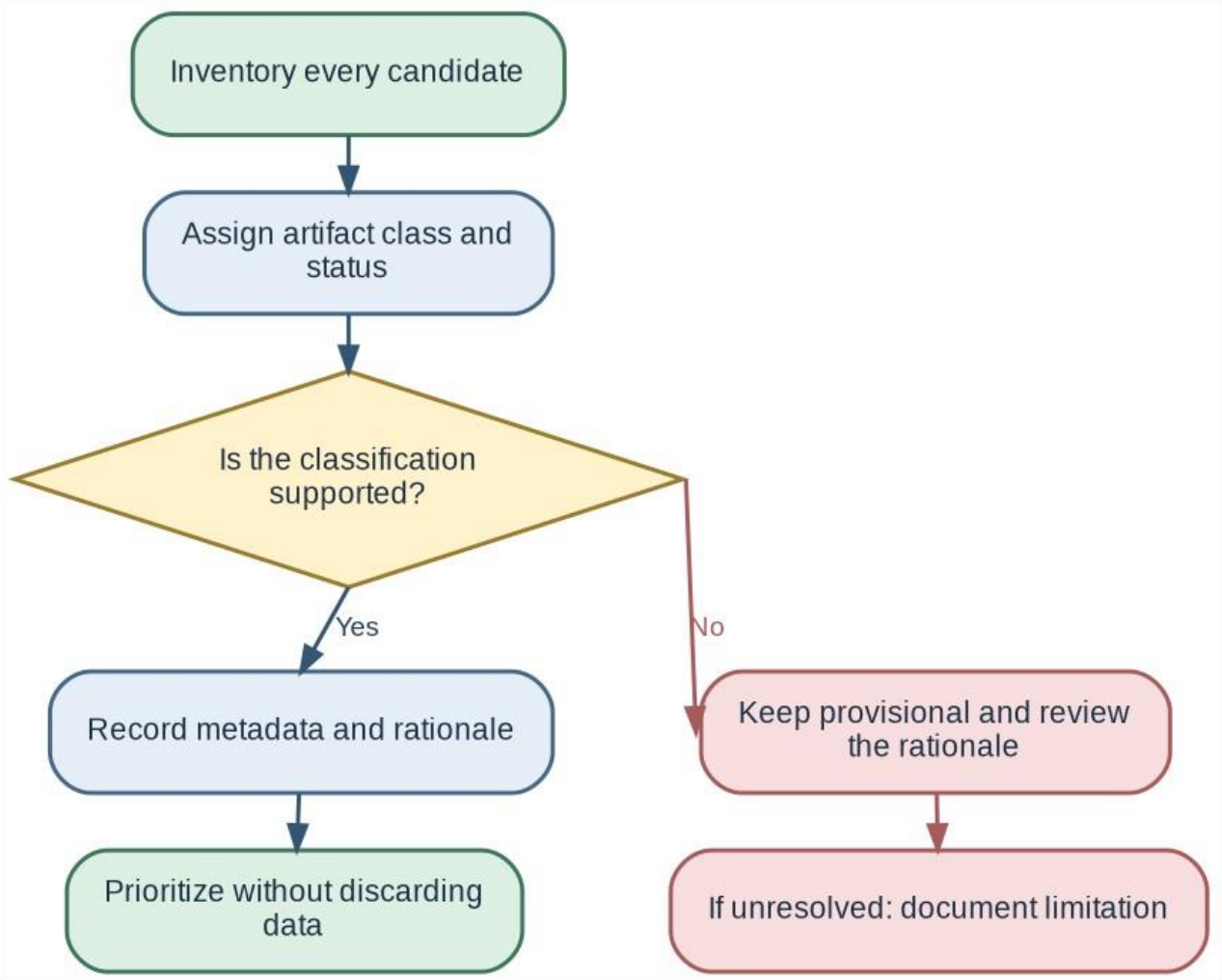
Flowchart 23. Question-specific reasoning sequence.

24

Prepare a classification scheme for user-created files, system/application files, potentially suspicious files, deleted/recovered artifacts, and supporting/contextual evidence.

Use a classification scheme that captures artifact type and why it matters, rather than labeling every unusual file as malicious. User-created files include documents, spreadsheets, archives, and project files. System/application files include executables, configuration, caches, logs, and program databases. Potentially suspicious files include unknown executables, scripts, unusual archives, or files with suspicious names or hashes; these require verification.
Deleted/recovered artifacts include remnants from unallocated space or Volume Shadow Copies.
Supporting/contextual evidence includes logs, shortcuts, browser history, user-profile records, and timestamps.

For every item, record evidence ID, original or recovered path, source image/snapshot, size, timestamps, hash, status, classification rationale, relevance, and limitations. Assign confidence only after examination. The same file may have multiple roles; for example, a recovered document can be a user-created file and a deleted artifact.
Preserve all categories and explain why an artifact was selected for deeper analysis.



Flowchart 24. Question-specific reasoning sequence.

Evidence classification scheme

Class	Examples	Handling / interpretation
User-created files	Documents, spreadsheets, archives, project files.	Record path, owner/SID if available, hashes, version, and relevant content.
System/application files	Executables, configuration, logs, cache, application databases.	Interpret in relation to the OS/app version; preserve required context.

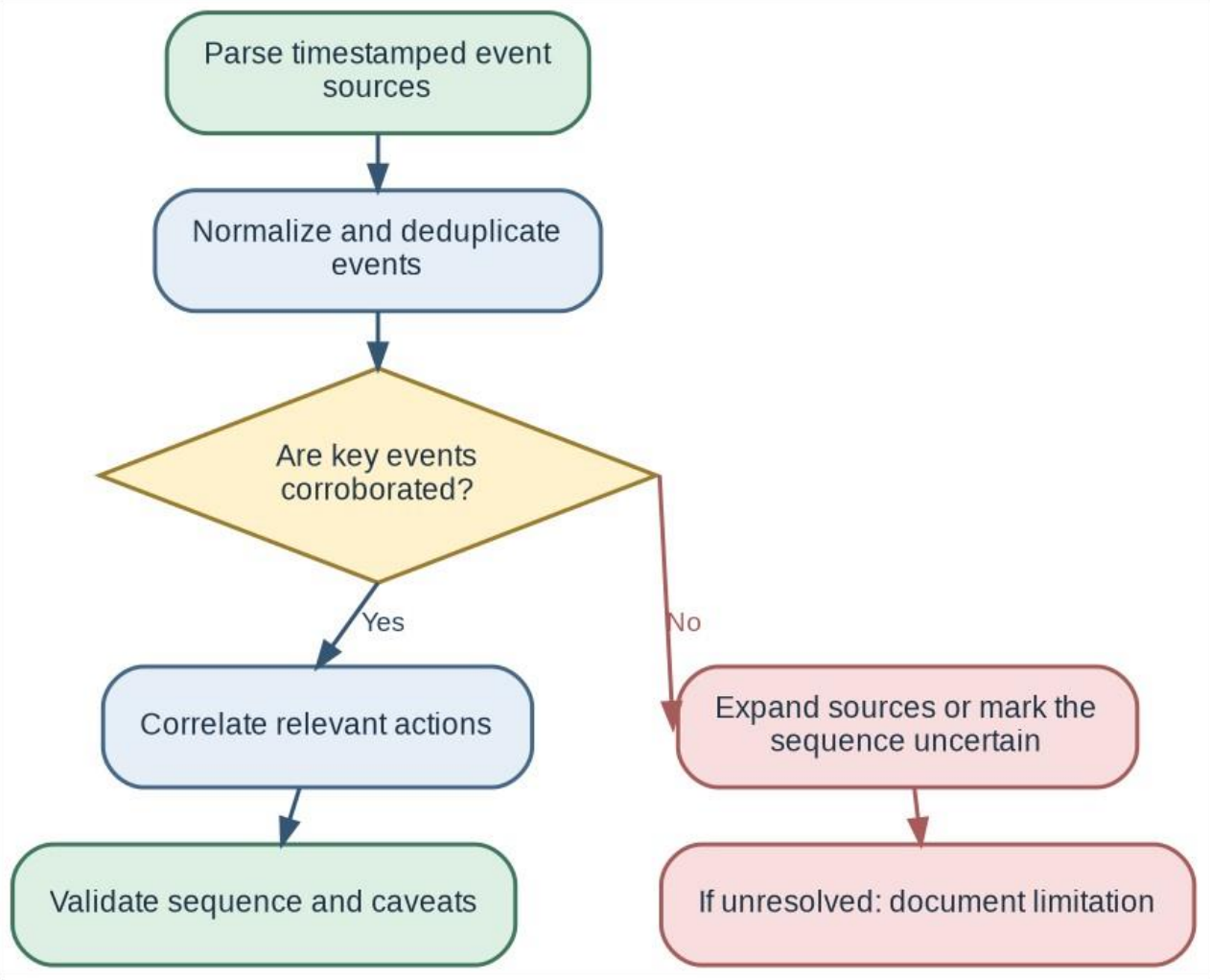
Class	Examples	Handling / interpretation
Potentially suspicious files	Unknown executable/script, unusual archive, file with suspicious hash/behavior.	Do not execute on the evidence host; validate hash and analyze in a controlled environment.
Deleted/recovered artifacts	Deleted directory entries, unallocated-space recovery, VSS versions.	Record recovery method, source snapshot/offset, integrity, and missing metadata.
Supporting/contextual evidence	Shortcuts, browser records, logon logs, user-profile or device history.	Use to corroborate; identify alternative causes and timestamp limits.

25

Explain how Event Analysis can assist in reconstructing a sequence of activities.

Event Analysis extracts and organizes events from different artifacts into a common timeline. It can show, for example, an interactive logon, process execution, USB insertion, file modification, browser download, file deletion, or snapshot creation. Related Windows Event Logs, registry data, file-system metadata, shortcut files, application records, and security logs can provide complementary evidence about an action and its context.

The examiner should normalize time zones, distinguish raw timestamp fields, deduplicate repeated events, and label each event with source and confidence. Events need not be complete and may be missing because of log rotation, auditing configuration, or retention limits. A timeline establishes sequence and possible relationships; it does not automatically prove causation or human intent. Important events should be checked against independent artifacts and reported with gaps and alternate interpretations.



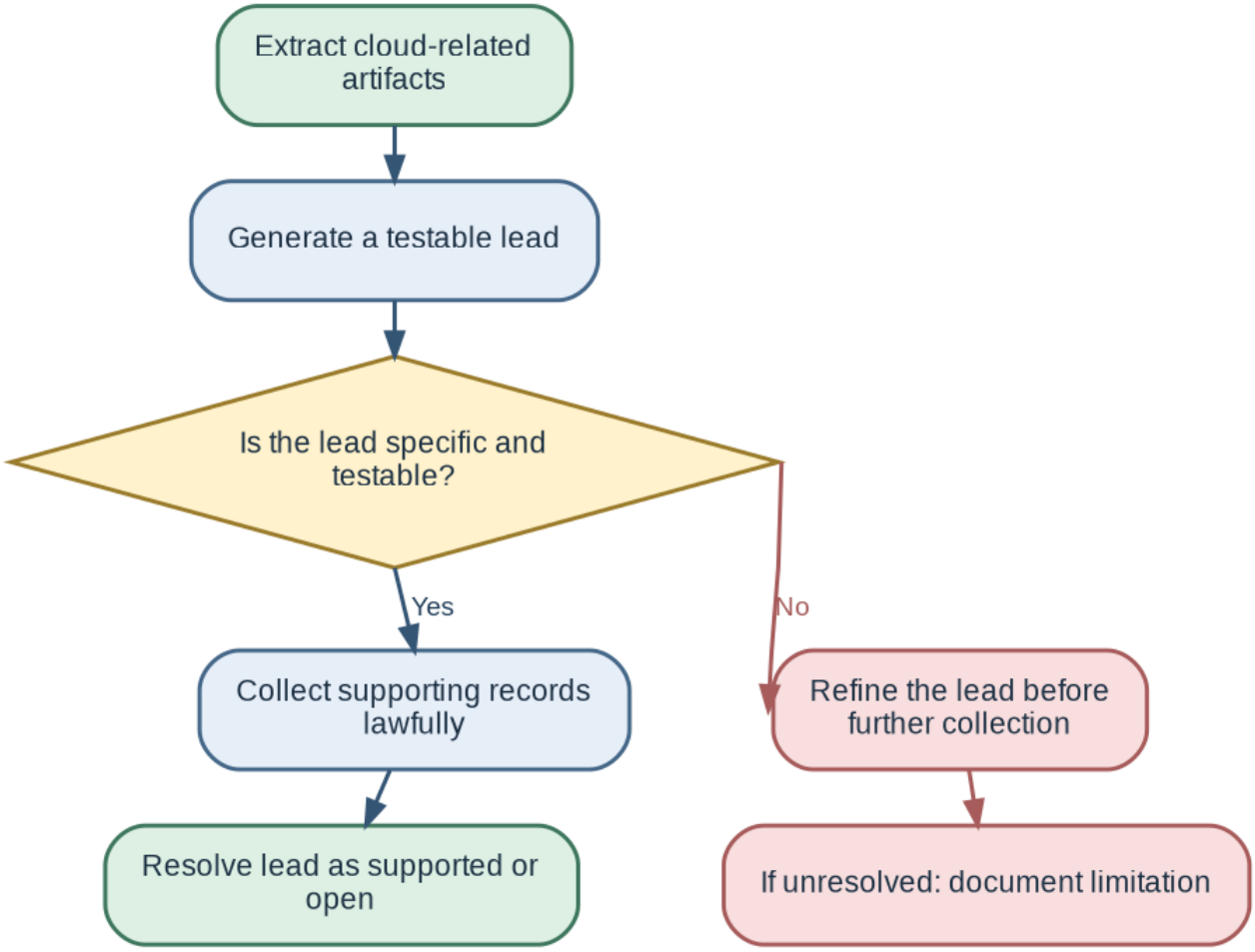
Flowchart 25. Question-specific reasoning sequence.

26

Explain the purpose of Cloud Analysis and Lead Analysis in an investigation and identify the kinds of questions each may help answer.

Cloud Analysis considers evidence associated with online storage or cloud-connected applications: sync logs, account records, audit events, object versions, shared links, access IPs, download history, and locally cached references. It may help answer which account accessed a resource, whether a file was synced or shared, whether a version existed at a particular time, and which provider-side records should be requested. Local artifacts must be distinguished from provider-side records and from assumptions about remote state.

Lead Analysis uses a finding to generate a targeted investigative question or next collection step. For example, a file reference in browser history may lead investigators to a download record; a cloud-sync path may lead to authorized account audit records; a hash may lead to a known-file comparison. A lead is not a conclusion. The investigator records why it was pursued, obtains the required authority, tests it against independent sources, and closes it as supported, unsupported, or unresolved. Exact feature labels depend on the forensic tool and version.



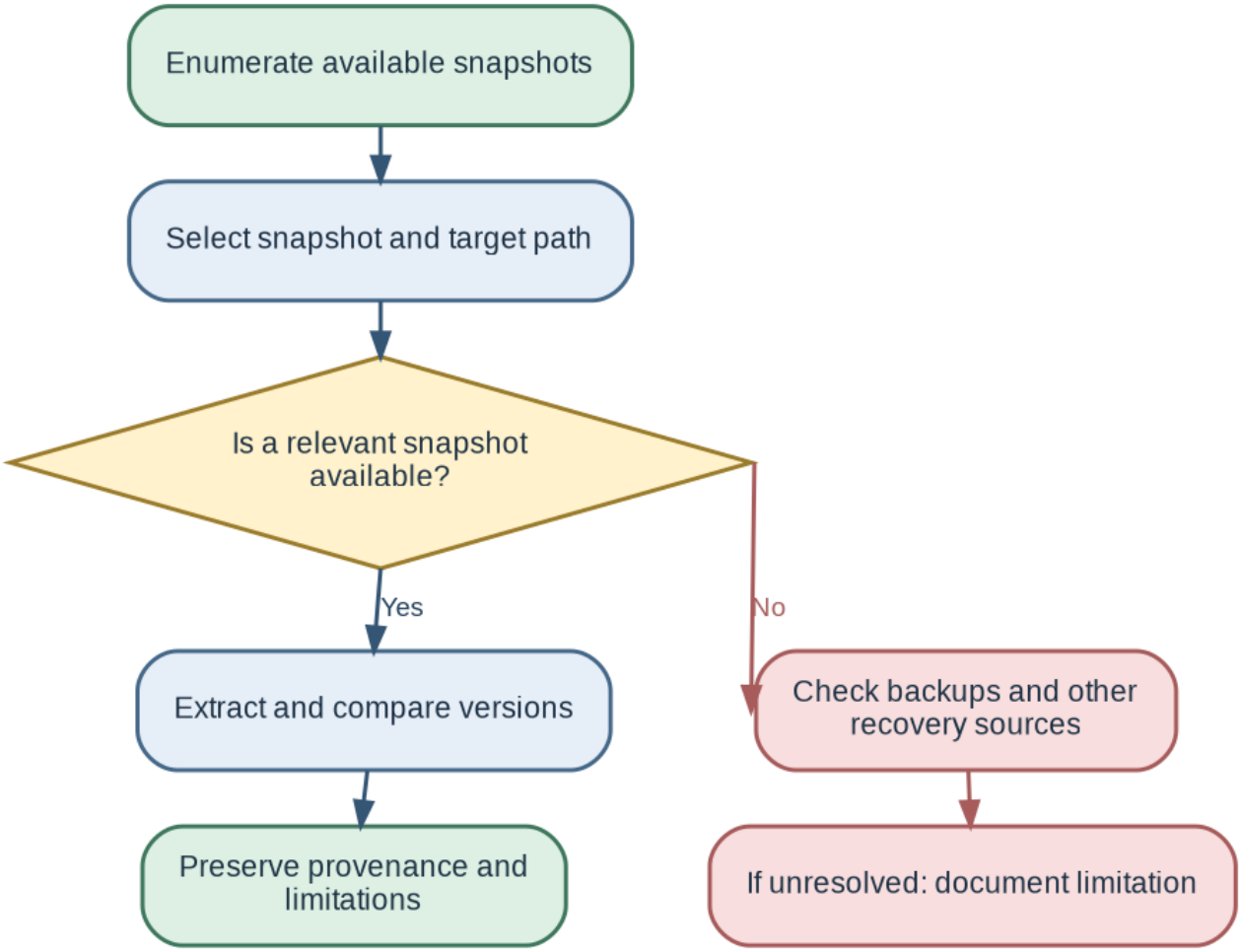
Flowchart 26. Question-specific reasoning sequence.

27

Describe how Volume Shadow Copy analysis can help recover or compare earlier versions of evidence.

Windows Volume Shadow Copy Service can maintain point-in-time snapshots used by system restore, backup, or applications. When a snapshot predates a deletion or change and the relevant data were retained, an examiner may compare a prior file version with the current state or recover an earlier copy. The comparison can reveal content changes, prior paths, earlier metadata, or the existence of a file that is absent from the current file-system view.

Work from a verified forensic image and enumerate available snapshots before browsing them. Record each snapshot's identifier, creation time, source volume, and acquisition context. Extract prior versions to a separate working area, preserve their source paths and snapshot identifiers, and calculate hashes. Snapshots are not complete historical archives: retention, overwritten blocks, file-system conditions, and snapshot configuration may limit recovery. A prior version proves that data existed in that snapshot, not automatically who created or deleted it.



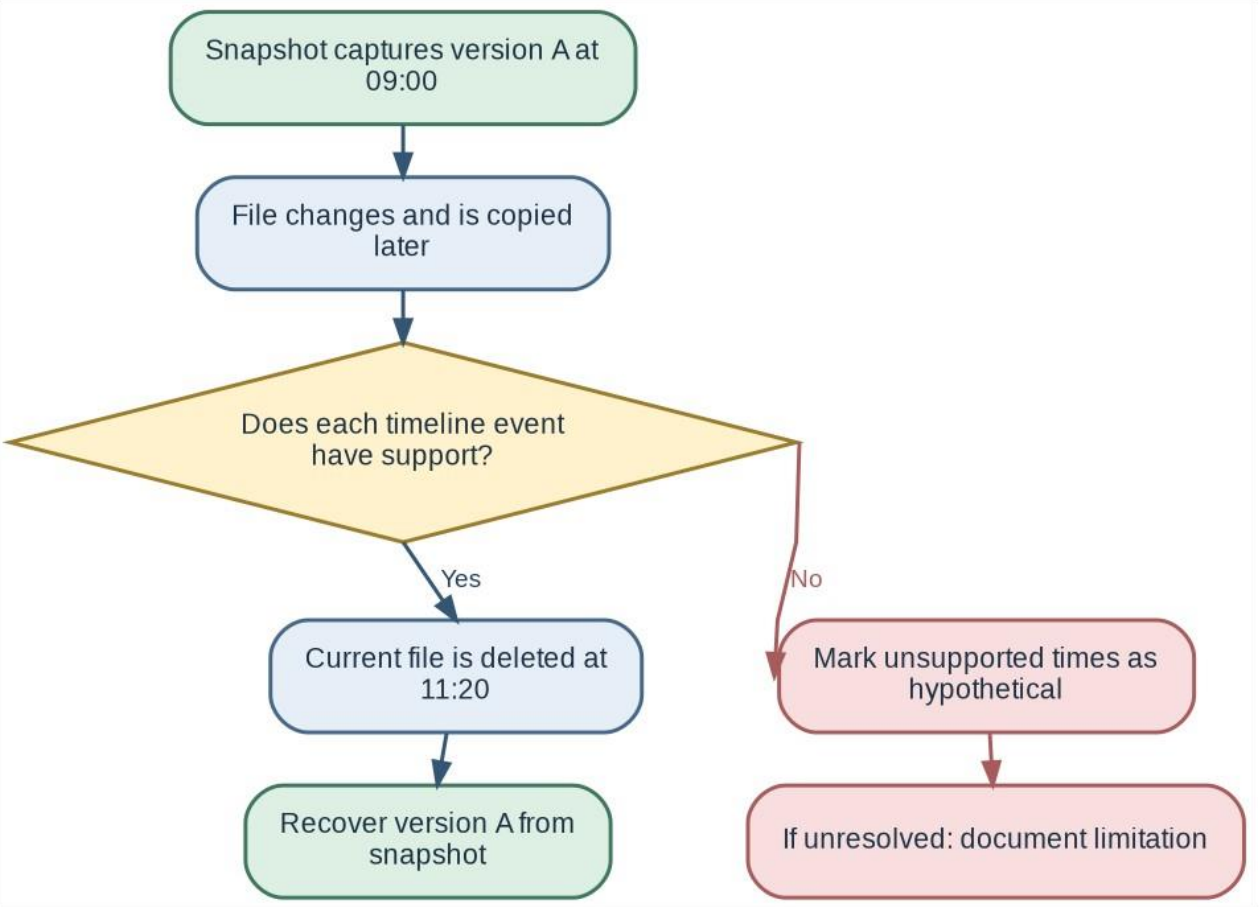
Flowchart 27. Question-specific reasoning sequence.

28

Construct a hypothetical timeline showing how a file could be created, modified, copied, deleted, and later recovered from a Volume Shadow Copy.

The following is a hypothetical UTC-normalized example, not an observation from the provided case. At 09:00 a Volume Shadow Copy is created and contains `ExamPlan.docx` version A. At 10:15 user account U creates or saves version B. At 10:42 the file is copied to a USB path. At 11:05 the local file is modified again to version C. At 11:20 the file is deleted from the current volume. During later examination, the 09:00 snapshot is found to contain version A, which is extracted from a verified forensic copy.

The timeline must label the source for each event: snapshot metadata, file-system entry, USB records, application metadata, or copy operation. The sequence is only supported if the artifacts align and the timestamp semantics are understood. The 09:00 snapshot would not contain changes that occurred later. A matching hash can establish identity between recovered bytes and another preserved copy, but it does not prove the same content was copied at 10:42 unless independent evidence supports that connection.



Flowchart 28. Question-specific reasoning sequence.

Hypothetical event timeline (UTC; illustrative only)

Time (UTC)	Hypothetical event	Potential source	Interpretive caution
09:00	VSS snapshot records ExamPlan.docx version A.	Snapshot metadata and extracted version.	Shows version A existed in this snapshot, not who created it.
10:15	Account U saves modified version B.	File-system/application artifacts.	Confirm the field and user/session context; time may be skewed.
10:42	A copy is written to USB.	USB artifact, file hash, file-system records.	A matching file alone does not establish the exact copying time.
11:05	Local document is modified to version C.	Current filesystem and application artifacts.	Copying or sync may affect timestamps.

Time (UTC)	Hypothetical event	Potential source	Interpretive caution
11:20	Current-volume file entry is deleted.	Filesystem / USN records if available.	Deletion may be routine or application-driven; intent remains unproven.
Later examination	Version A is recovered from the 09:00 snapshot.	Verified forensic image and VSS extraction log.	Preserve snapshot ID, extraction provenance, and output hash.

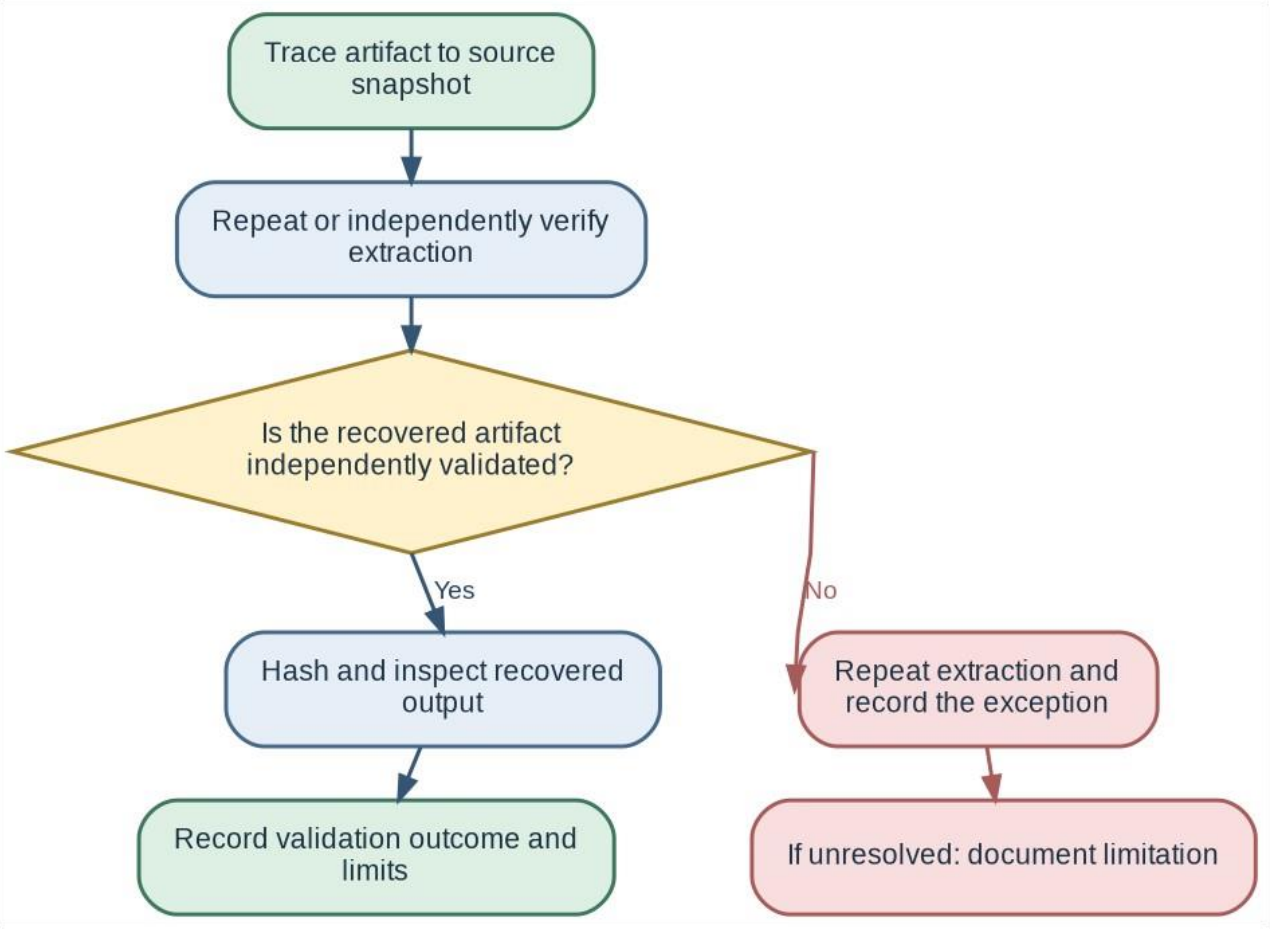
29

Explain how the recovered artifact should be validated before it is included as a finding in the final report.

Validation starts with provenance: confirm the image hash, snapshot identifier, source volume, target path, and extraction method. Re-extract the artifact from a separate working copy or repeat the extraction using an independent method when possible. Calculate a hash for the recovered output, compare it with any known reference copy, and use a format-aware parser to confirm that the file opens and its internal structure is plausible. Preserve the original recovered bytes even if the file is corrupted or incomplete.

Check whether timestamps came from the snapshot, the current file system, or the extraction process. Inspect metadata and content for consistency with related artifacts; compare with logs, backups, or another source.

Record the tool and version, parameters, warnings, errors, and validation outcome. A file that opens successfully is not automatically authentic, and a failed open does not prove the content is irrelevant. Clearly state what was validated and what remains uncertain.



Flowchart 29. Question-specific reasoning sequence.

Recovered-artifact validation checklist

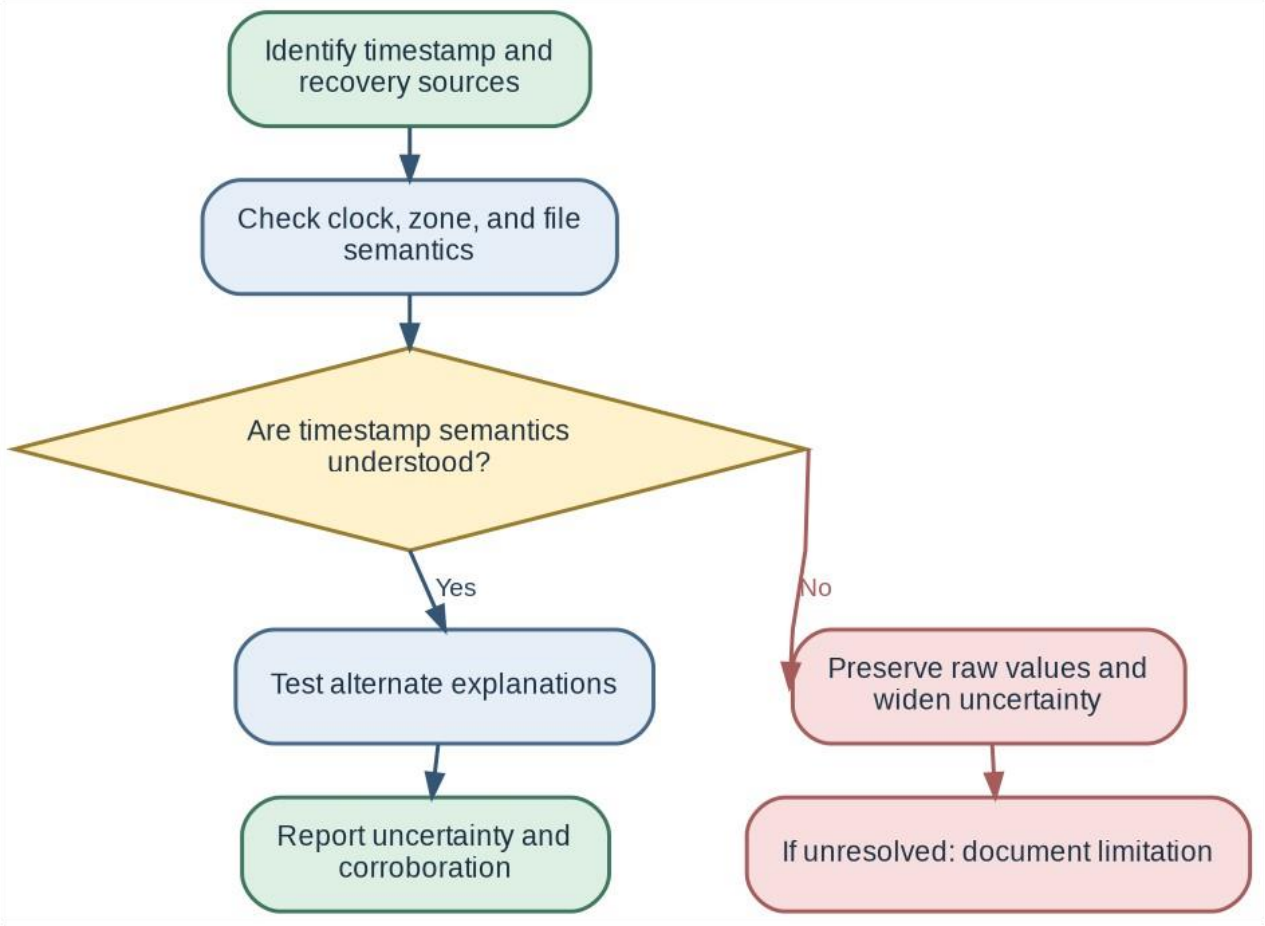
Check	What to confirm	Record
Provenance	Verified image hash, volume, snapshot ID, original path, offset if applicable.	[Result / source ID]
Repeatability	Repeat extraction or compare an independent method where practical.	[Tool/version and result]
Integrity	Hash recovered output; compare with reference copies if available.	[Algorithm and digest]
Content and format	Use a format-aware parser; inspect completeness and internal structure.	[Observations and warnings]
Context	Compare metadata and content with logs, backups, other devices, or versions.	[Corroboration / conflicts]
Reporting	State source, technique, uncertainty, missing metadata, and alternative explanations.	[Finding ID and conclusion]

30

Identify possible sources of error or alternative explanations that could affect the interpretation of timestamps and deleted artifacts.

Timestamp interpretation may be affected by incorrect system clocks, time-zone conversion errors, daylight-saving adjustments, differing timestamp semantics, copied files retaining old modified times, or application-level updates. Snapshot creation time can differ from the time a file was created or deleted. Logs may be missing, rotated, or generated on a different host. The examiner must also consider tool parsing errors and whether the recovered record came from the current volume, a snapshot, or unallocated space.

A deleted entry may reflect routine cleanup, application behavior, a user's normal workflow, malware, or deliberate concealment; deletion alone does not establish motive. A recovered artifact could be a duplicate, an incomplete fragment, or a file synchronized from another device. Investigators should preserve raw values, document the conversion method and clock assumptions, corroborate critical events, and avoid forcing events into a precise order when the uncertainty intervals overlap.



Flowchart 30. Question-specific reasoning sequence.

Assignment 2(5) - Multi-Source Investigation of a Suspected Cyber Incident

Questions 31-38

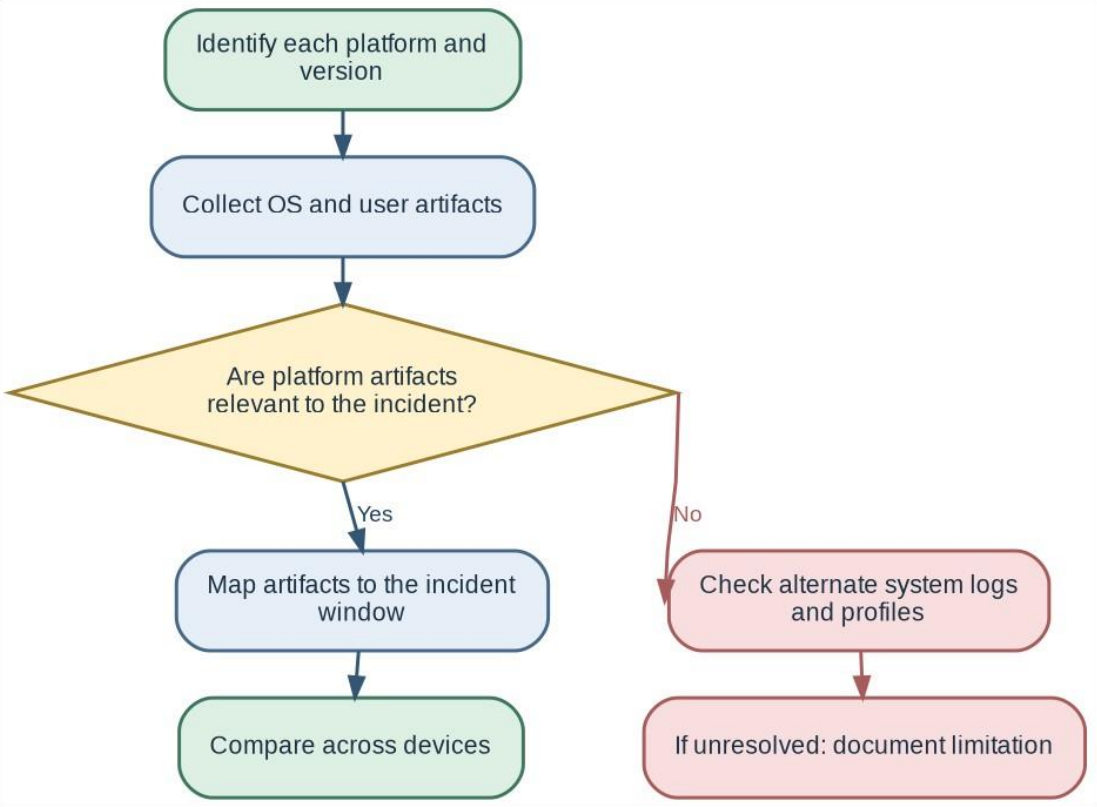
A student account may have been used to access an online service without authorization. Potential sources include a Windows computer, browser artifacts, email and messaging records, a mobile phone, and Internet/cloud records. The analysis correlates sources while observing the relevant authority, privacy, and retention limits.

31

Identify the potential evidence available from Windows and other operating systems in this case.

On Windows, useful evidence may include Security and System Event Logs, registry hives, NTFS metadata, LNK files and Jump Lists, Prefetch or other execution artifacts where enabled, browser profiles, installed applications, scheduled tasks, device-connection history, DNS cache if available, and endpoint security logs. Evidence may also exist in RAM or active network sessions if timely, authorized live collection is warranted.

On Linux, relevant sources can include authentication logs, systemd journal, shell history, service logs, package-manager records, file-system metadata, cron jobs, browser profiles, and SSH configuration or logs. On macOS, sources can include Unified Logs, FSEvents, Spotlight metadata, plist preferences, browser data, application containers, and Time Machine artifacts. Artifact availability varies by OS version, configuration, retention, permissions, and user activity. Record the platform and version, and avoid interpreting the absence of an artifact as proof that the activity did not occur.



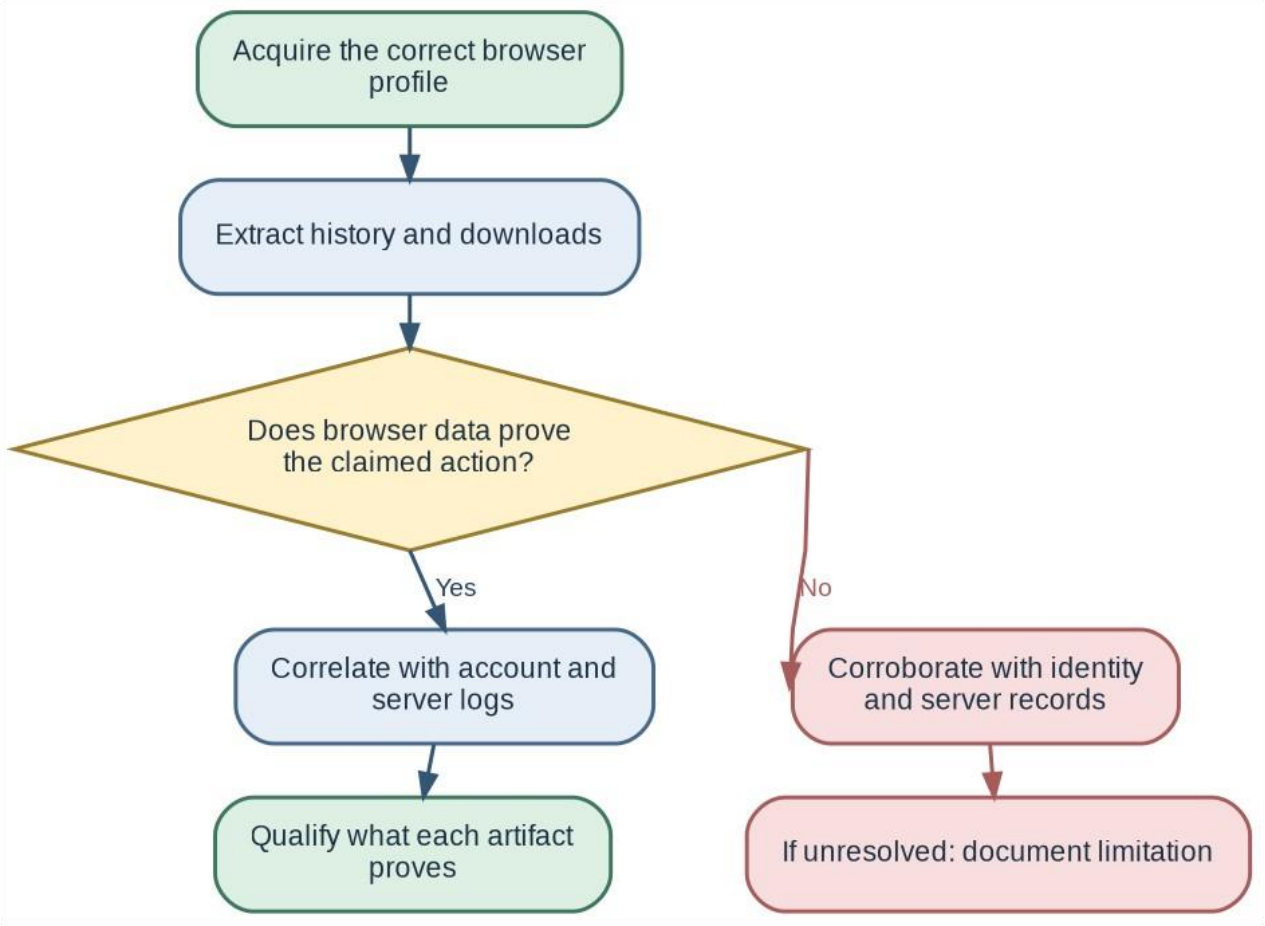
Flowchart 31. Question-specific reasoning sequence.

32

Explain what browser artifacts may reveal about user activity and why browser evidence must be interpreted carefully.

Browser artifacts may include history, typed URLs, search terms, cached pages, cookies, download records, session restore data, bookmarks, form data, extensions, service workers, and profile or synchronization information. Download records may connect a URL with a local file and a timestamp; cache or history entries may provide context for a phishing page, authentication attempt, or online-service access.

Interpretation requires caution. A browser may prefetch or load resources in the background, extensions may generate traffic, pages may be opened by redirects, and a shared or synchronized profile may contain activity from another device. Private-browsing modes and retention settings change what remains. A history entry usually shows that the browser recorded a URL, not that the user read or understood the page or successfully completed an action. Correlate browser data with server-side logs, authentication events, downloaded-file hashes, and the relevant account context.



Flowchart 32. Question-specific reasoning sequence.

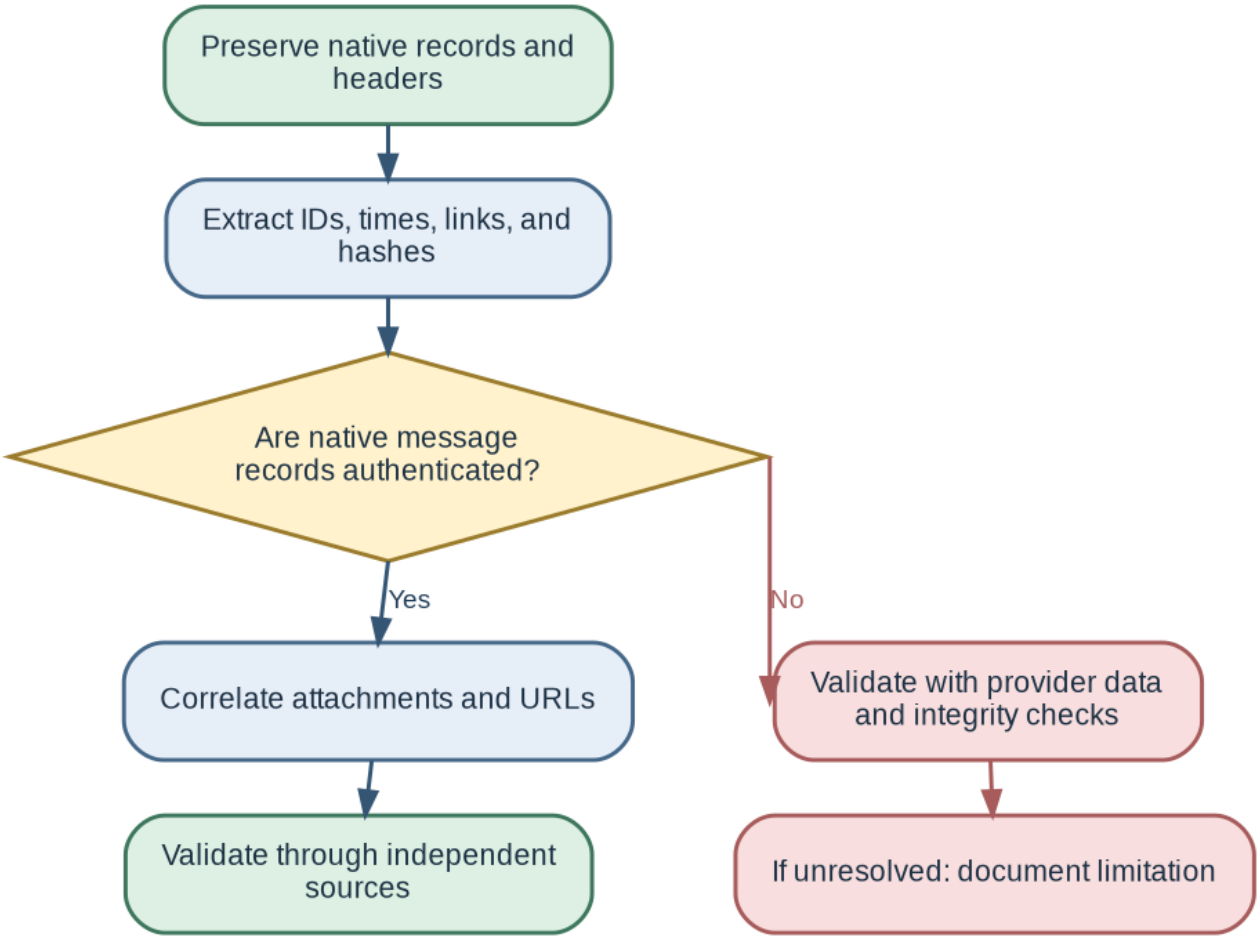
33

List the types of evidence that may be available from email and messaging systems and explain how they can be correlated with other artifacts.

Email evidence can include full headers, sender and recipient fields, message IDs, received-server hops, timestamps, attachments, message bodies, mailbox rules, login alerts, provider audit records, and local mail databases. Messaging evidence may include conversation databases, contact identifiers, delivery/read markers, message timestamps, attachment paths, application logs, notifications, and cloud backups where lawfully available. Screenshots alone are weaker than preserved underlying records because they can be cropped or altered.

Correlate a message URL with browser history, an attachment hash with a downloaded file, a login alert with authentication logs, and a message time with mobile-device activity. Preserve original exports and header data, record the acquisition method and time zone, and distinguish server timestamps from device display times.

Messaging platforms may use end-to-end encryption or short retention, so content might not be available from provider records. The absence of a message in one source does not prove it was never sent.



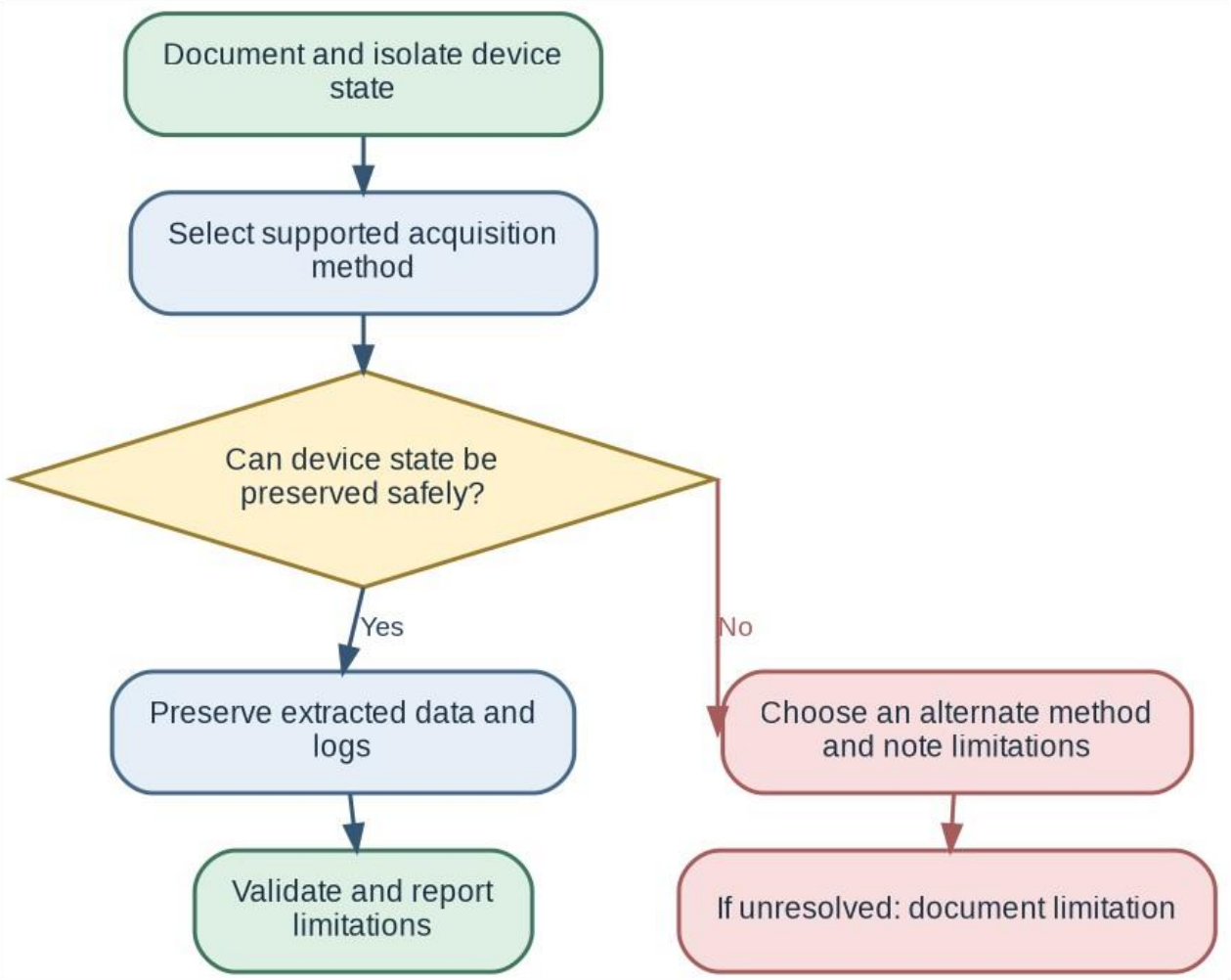
Flowchart 33. Question-specific reasoning sequence.

34

Explain the forensic considerations when examining a mobile phone and identify challenges associated with mobile evidence.

Document the handset make/model, serial or IMEI as legally appropriate, SIM and memory-card details, lock and power state, visible notifications, network connections, date/time, and physical condition. Obtain explicit authority for the device and data categories. A qualified examiner should choose logical, file-system, or physical acquisition based on the device, tool support, and investigative need. Avoid unnecessary interaction: unlocking, rebooting, updating, or connecting a device may change data or make encrypted data inaccessible.

Mobile challenges include strong encryption, lock screens, hardware-backed keys, changing OS versions, app sandboxing, short-lived tokens, remote wipe, cloud synchronization, fragmented databases, and limited tool support. Isolate radios or network access using a documented method appropriate to the device and risk; a Faraday enclosure can help but must be checked for signal leakage and power considerations. Preserve extracted databases, logs, acquisition reports, and hashes. NIST SP 800-101 Rev. 1 discusses preservation, acquisition, examination, analysis, and reporting of mobile evidence [2].



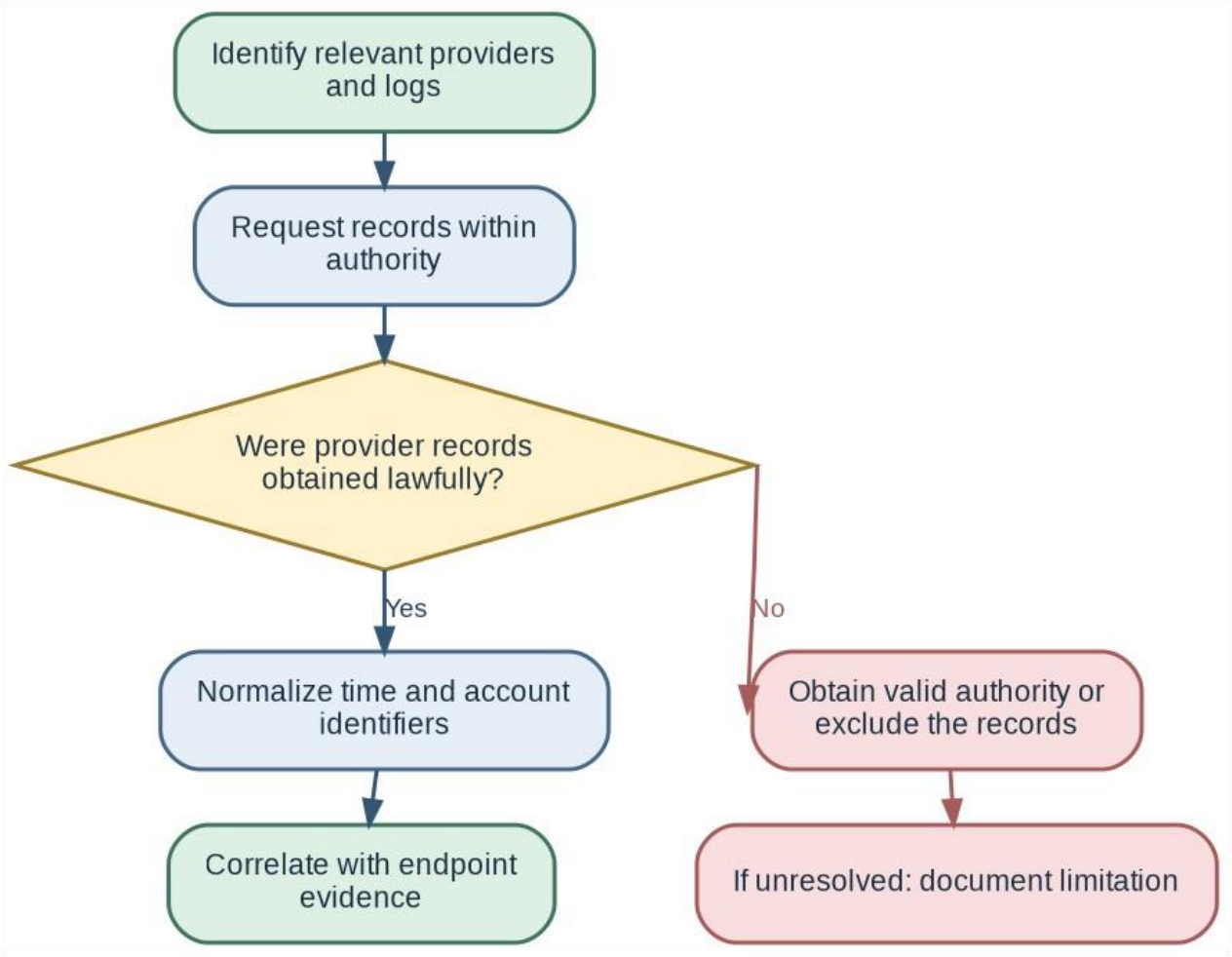
Flowchart 34. Question-specific reasoning sequence.

35

Discuss Internet and cloud-based evidence sources that may be relevant to the investigation.

Internet evidence may include DNS resolver logs, DHCP leases, firewall and proxy records, VPN logs, network flow records, intrusion detection alerts, and ISP connection records where available and properly requested. Cloud sources may include identity-provider sign-in logs, application audit trails, object-storage access logs, file version history, sharing permissions, API activity, backup histories, and synchronization events. Provider records can distinguish a local file reference from an authenticated cloud action.

Record the source organization, account and tenant identifiers, time range, time zone, retention window, collection method, and integrity information. Preserve original exports and note any transformation by the provider or portal. IP addresses do not uniquely identify a person or physical location: NAT, VPNs, shared Wi-Fi, proxies, and mobile networks complicate attribution. Cloud data may cross jurisdictions or be retained for a limited time, so legal process and timely preservation requests may be necessary.



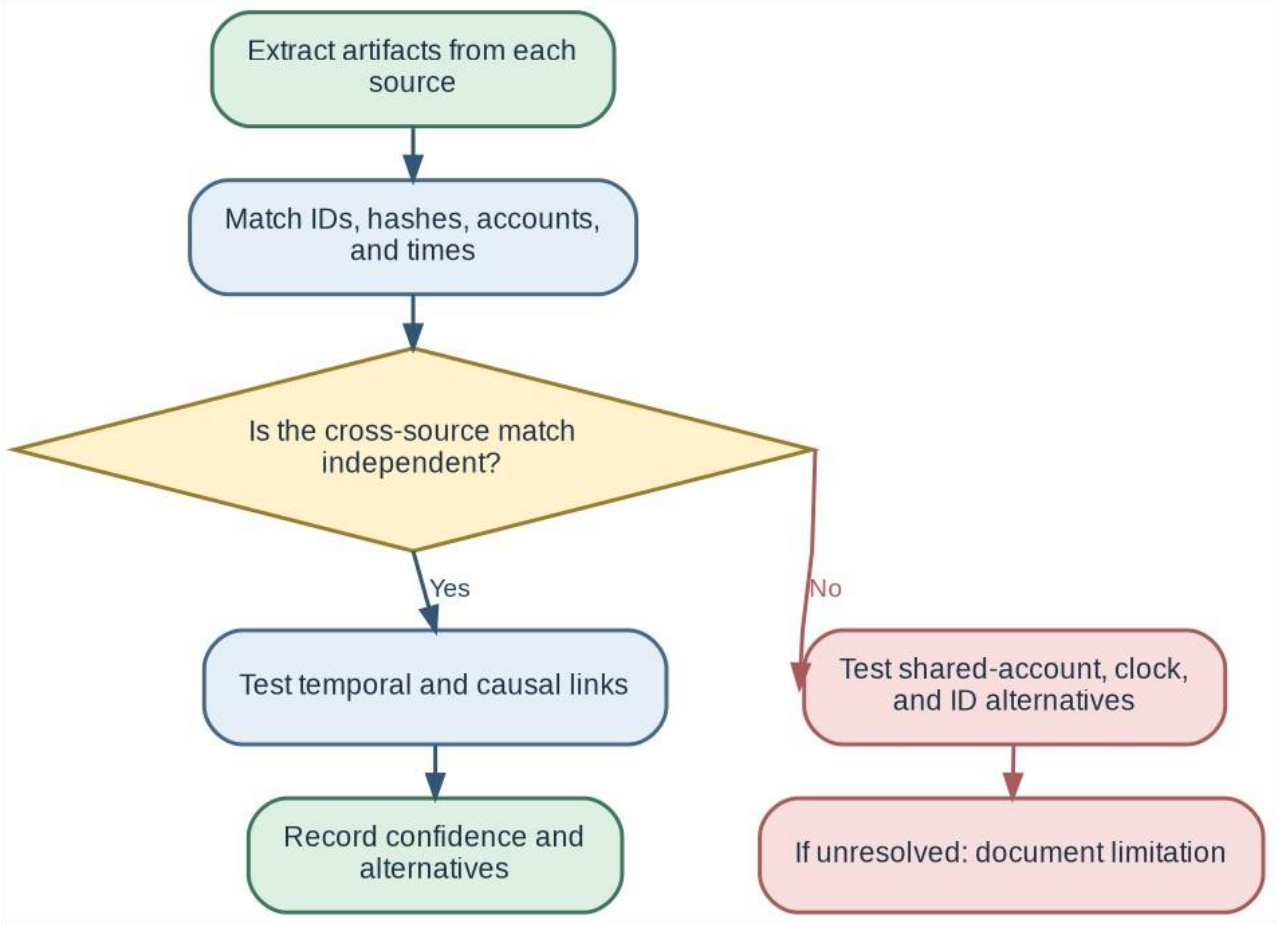
Flowchart 35. Question-specific reasoning sequence.

36

Prepare a cross-source correlation table linking at least five hypothetical artifacts from the computer, browser, email/messaging, mobile, and cloud sources.

A cross-source matrix should align artifacts around a shared event, not simply place unrelated records in one table. The hypothetical rows below illustrate how an email link, mobile message, browser navigation, Windows logon, and cloud audit entry might relate in time. The example is synthetic and does not imply that these events occurred in the supplied case. Use exact IDs and timestamps from actual collected records when preparing a real report.

For each relationship, specify the common key (account, session ID, URL, IP, file hash, device ID, or time window), the degree of agreement, and an alternative explanation. Preserve the original source records so the correlation can be reproduced. A sequence of correlated artifacts may support that an account was used but still leave unresolved whether the account holder personally operated the device or whether credentials were compromised.



Flowchart 36. Question-specific reasoning sequence.

Cross-source correlation matrix (hypothetical)

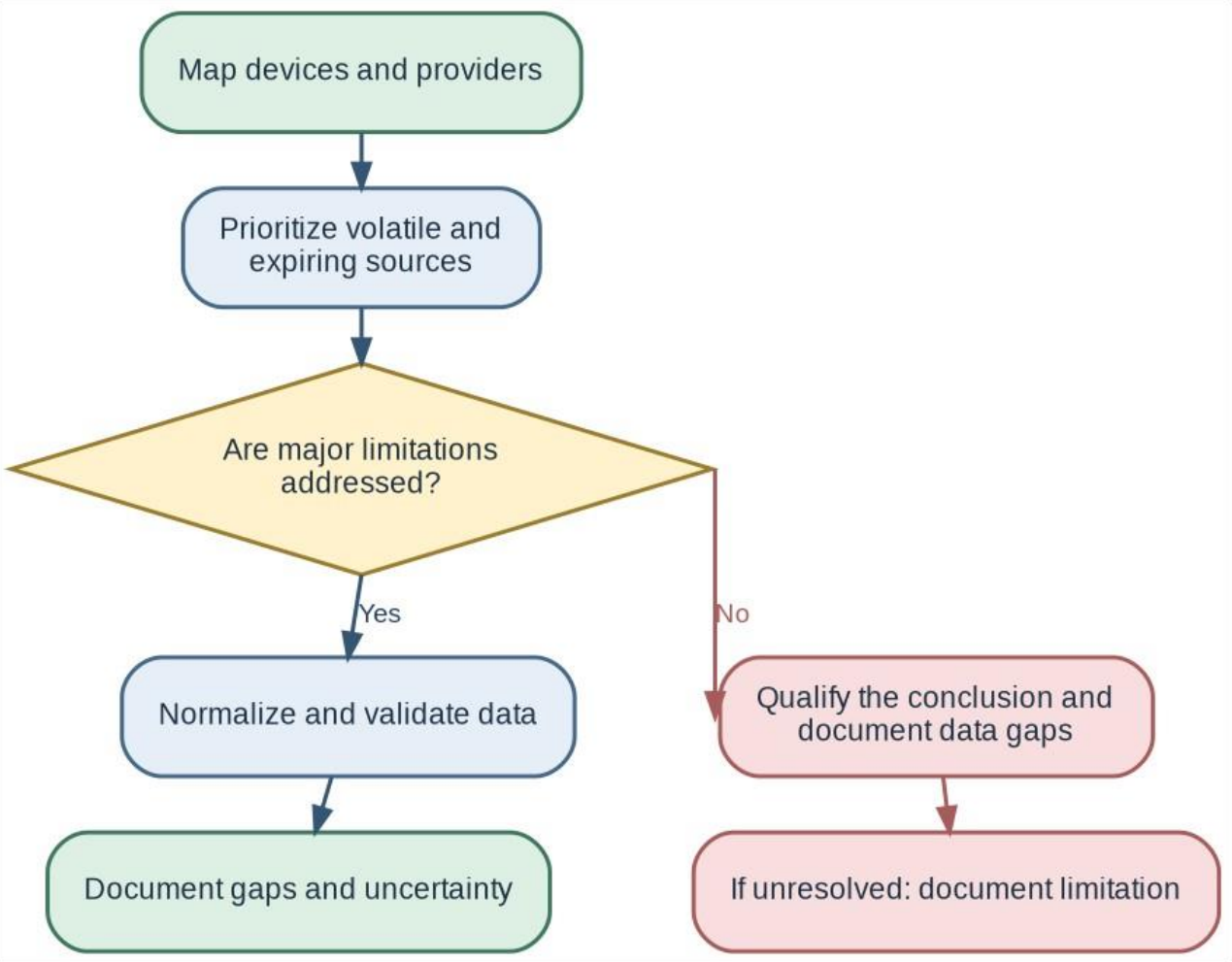
ID / source	Hypothetical artifact	Shared correlation key	What it may support	Limitation
A - Email	20:55 message contains a service sign-in URL.	URL, Message-ID, recipient account.	May explain how the link reached the account.	Receipt does not prove the recipient opened it.
B - Mobile	20:58 message notification records the same URL.	URL, device time, application account.	Corroborates delivery to a mobile context.	Notification may be delayed or synchronized.
C - Windows	21:10 OS log shows an account logon/session.	Account SID, session ID, normalized time.	Establishes an account session on the endpoint.	Does not identify the human operator by itself.
D - Browser	21:12 browser history records navigation to the URL.	URL, profile, time, download record.	Corroborates access from a browser profile.	Background requests/redirects and shared profiles are possible.
E - Cloud	21:14 audit log records access to a service resource.	Cloud account, event ID, IP/session, time.	May provide direct record of a service-side action.	Requires provider provenance and credential-compromise analysis.
F - Download	21:17 local file has a hash matching cloud-exported file.	SHA-256, file path, timestamp.	Supports identity of the compared content.	Hash match does not prove who downloaded it.

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Explain at least six major challenges in digital forensics when evidence is distributed across multiple devices and cloud services.

Distributed investigations face heterogeneous operating systems and application formats, encryption and locked devices, volatile data and short retention, high evidence volume, inconsistent clocks and time zones, proprietary or changing cloud APIs, jurisdiction and authorization boundaries, and privacy constraints. Artifacts may be synchronized, duplicated, cached, or transformed across devices, making it difficult to determine which copy is original or when a change occurred.

Other challenges include provider-side logs that are unavailable or incomplete, remote-wipe risks, anti-forensic actions, deleted or fragmented records, tool support gaps, and the difficulty of validating parsers for proprietary formats. Manage these issues with a documented scope, prioritized preservation, tool validation, standard time normalization, independent corroboration, and explicit confidence ratings. The report should disclose missing sources, retention gaps, and any conclusions that depend on unverified assumptions rather than conceal them.



Flowchart 37. Question-specific reasoning sequence.

Cross-source digital forensics challenges

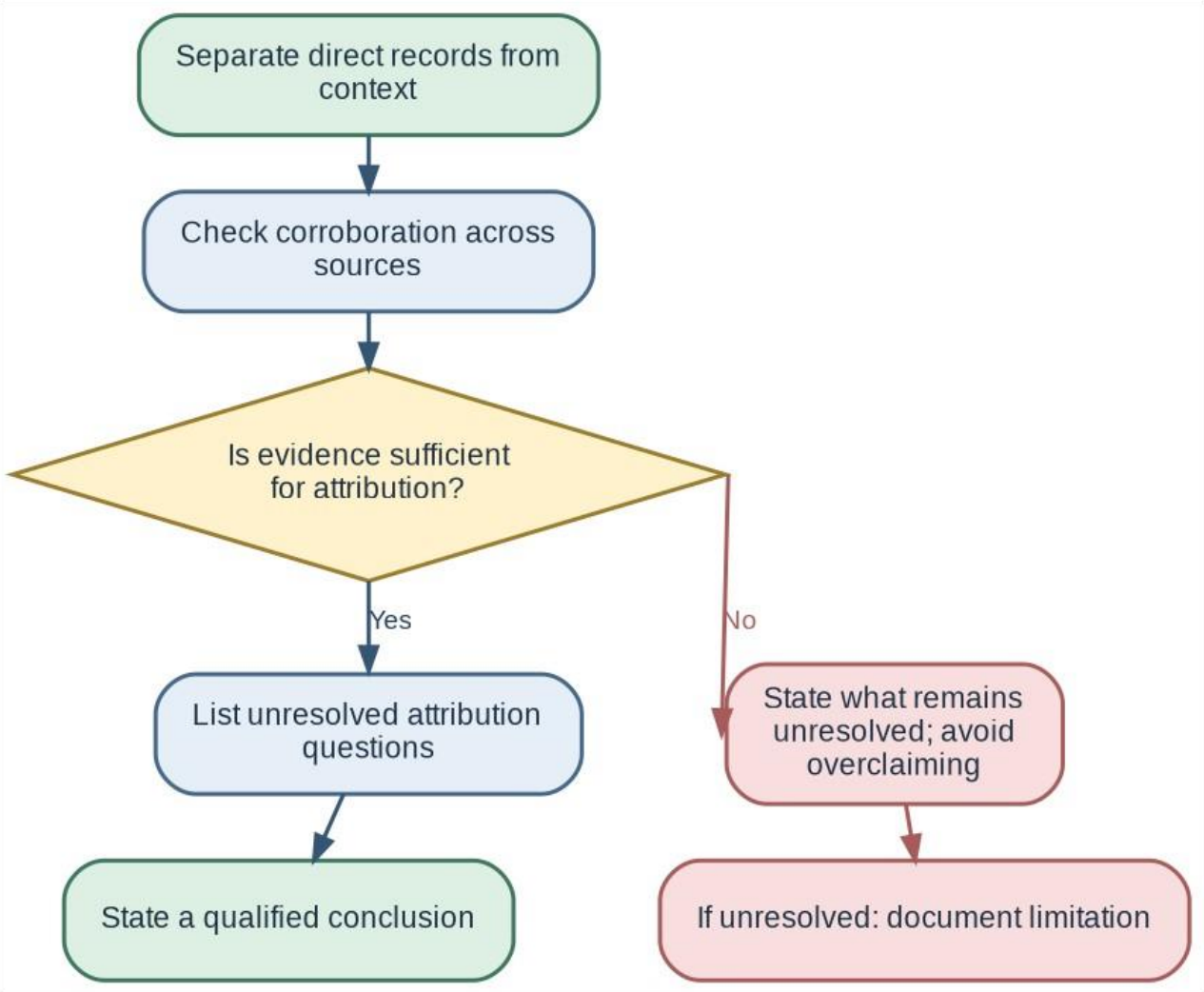
Challenge	Effect on investigation	Mitigation
Heterogeneous systems and formats	Artifacts have different structures and parser support.	Record versions; validate tools; preserve native exports.
Encryption and locked devices	Content or keys may be unavailable.	Preserve state; use authorized methods; document limits.
Volatility and retention	Logs/tokens may disappear or expire.	Prioritize lawful preservation and timely acquisition.
Clock and time-zone differences	Events may be misordered.	Preserve raw times; normalize separately; estimate skew.
Cloud jurisdiction and access	Records may require provider process or cross-border authorization.	Coordinate legal review and scoped preservation requests.
Scale and duplication	Large volumes and sync copies increase noise.	Prioritize, deduplicate carefully, preserve provenance.
Privacy and scope	Unrelated personal data may be exposed.	Apply minimization, access control, retention policy.
Tool and parser limits	Extraction errors or unsupported apps may distort results.	Validate key artifacts independently and report error logs.

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Prepare a final investigation conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Hypothetical conclusion: the available artifacts should be assessed in three categories. Direct evidence would include a verified cloud audit record or service-side event showing a defined operation by a specific authenticated account, linked to a session and time. Corroborating evidence could include a matching browser URL, a downloaded file with a matching hash, a related email, a mobile notification, or Windows login records. Each item supports the proposition only to the extent described by its source and reliability.

Unresolved questions may include whether the student personally controlled the session, whether credentials were stolen or shared, whether the device clock was accurate, whether logs are complete, and whether the provider record has been collected in a legally sufficient form. Without the actual artifacts and validation results, the proper conclusion is conditional: the scenario offers potential avenues for correlation but does not establish that the student committed the incident. A final investigative summary should state findings, confidence, limitations, and recommended follow-up without assigning guilt beyond the evidence.



Flowchart 38. Question-specific reasoning sequence.

Final conclusion: evidence categories

Category	Illustrative evidence	What can reasonably be concluded
Direct evidence	Verified service-side audit event showing a specific operation by an authenticated account/session.	Supports that the account/session performed the logged operation, subject to record integrity and identity limitations.
Corroborating evidence	Matching browser URL, Windows logon, mobile notification, related email, and file hash.	Can strengthen temporal/account linkage when independently sourced and consistent.
Unresolved questions	Who physically controlled the session; whether credentials were compromised; clock accuracy; completeness and legal sufficiency of records.	Must remain open until independently supported; no attribution of guilt is justified from hypothetical artifacts.

Selected References

[1] Kent, K., Chevalier, S., Grance, T., and Dang, H. (2006). Guide to Integrating Forensic Techniques into Incident Response. NIST Special Publication 800-86. https://doi.org/10.6028/NIST.SP.800-86

[2] Ayers, R., Brothers, S., and Jansen, W. (2014). Guidelines on Mobile Device Forensics. NIST Special Publication 800-101 Rev. 1. https://doi.org/10.6028/NIST.SP.800-101r1

[3] National Institute of Justice, Office of Justice Programs. Test Results for Digital Data Acquisition Tool: IXimager (Version 2.0, February 1, 2006). https://www.ojp.gov/pdffiles1/nij/217678.pdf

[4] Government of India, India Code. The Bharatiya Sakshya Adhiniyam, 2023 (Act 47 of 2023), effective 1 July 2024. https://www.indiacode.nic.in/indiacode/handle/123456789/20063?col=123456789%2F1362&view_type=search

[5] Boddington, R. (2016). Practical Digital Forensics. Packt Publishing. The assignment's IXImager/ASB terminology follows the supplied syllabus context; tool-specific behavior should be checked against the version actually used.

Abbreviations

Common terms

Term	Meaning
ASB	IXImager/ILookIX-associated evidence container format used in the supplied course context.
Hash	Fixed-length cryptographic digest computed over a specified byte sequence.
VSS	Volume Shadow Copy Service; Windows mechanism for point-in-time volume snapshots.
MFT	Master File Table in NTFS.
USN Journal	NTFS change journal recording file-system change events.
UTC	Coordinated Universal Time, used here as a common timeline reference.
Chain of custody	Chronological record of evidence possession, transfer, storage, and handling.